IVN2Eth Capture Module:

CM SerDes GMSL2/3

USER MANUAL

October 2025

Document version: 1.8 Firmware: v010.006 Product version: 3.0

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1. Introduction

1.1 About this document

This user manual contains important information for the proper and safe operation of Technica Engineering ‘s Capture Module Serdes GMSL2/3.

Read this user manual carefully before using the device for the first time and follow the instructions, paying special attention to the safety-related parts.

For the latest version of this document please contact Technica Engineering via Technical Sales.

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1.1.1 Scope

This document includes instructions about how to use Technica Engineering’s Capture Module Serdes GMSL2/3 properly, together with information about some of the complex use cases that can be covered by this device.

Troubleshooting is not part of the scope of this document. For support queries, please contact Technica Engineering via Support.

Please note that, due to the dynamic development of the Capture Module Serdes GMSL2/3, the information in this document is subject to change without prior notice to adapt to technical changes and updates. Some features may not be described yet in the user manual or may be described incompletely.

Make sure to always use the correct version of this document, matching the firmware and hardware versions in operation (check next section).

1.1.2 Validity

This user manual applies to CM Serdes GMSL devices with:

- Firmware version: v010.006

1.1.3 Glossary

Term	Definition / Explanation
DUT	Device Under Test
Eth	Ethernet
FPGA	Field Programmable Gate Array. A programmable logic device that can have its internal configuration set by firmware
gPTP	Generic Precision Time Protocol, IEEE 802.1AS
HW	Hardware
IVN	In-Vehicle Network
K26	Xilinx® System-on-Module containing FPGA SoC
MAX10	Intel® FPGA device
PPS	Pulse per Second
PLP	Probe Logger Protocol
PTP	Precision Time Protocol, IEEE 1588
TECMP	Technically Enhanced Capture Module Protocol, used for encapsulation of captured in-vehicle network data
ToD	Time of Day.
API	Application Programming Interface

Table 1-1: Glossary

Additionally, the following symbols are used throughout the document to highlight important information:



Information: Useful tips, hints, or background information.



Warning: Important information to consider for the correct use of the device.

2. Safety and warranty

2.1 General operating and safety strategy

Technica Engineering's products are designed for operation in automotive systems and supply voltages of a nominal 12 V or 24 V. The applicable limit, values adhere to the standard norms for 12 V or 24 V automotive onboard power systems correspondingly and can be found in the mentioned norms.

Should Technica Engineering products be operated in voltage ranges beyond those specified in the norms, which represents a breach of the conditions of operation, then this will void the product warranty and Technica Engineering will assume no liability whatsoever for the results and/or consequences thereof.

This is especially valid whenever the voltage level reaches or exceeds the limits of the low voltage directive. In this case, damage to the devices cannot be excluded. Due to the manufacturing characteristics of the devices, there is no imminent fire hazard from the device itself, if the devices are being operated in an environment according to the conditions of use. A secondary fire hazard cannot be excluded, should those conditions not be met. Protection against overvoltage cannot be provided in such a breach of the conditions of use.

2.2 General design rules for the power supply

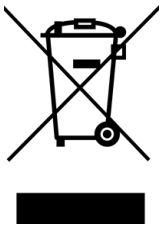
The power supply solution of Technica Engineering's products is equipped with a self-protection mechanism. This automatic function protects the internal electronic components against excessive temperature and too high a supply voltage by switching the device off. This automatic switch-off function is independent of any software function. Excessive temperatures in the device can be caused by high environmental temperatures or internal failures of the device. The device can protect itself by shutting down its power supply avoiding further damage to itself.

The operator/user of these devices must take measures for the right protection of the power supply lines and the according connector. The maximum power consumption of the Technica Engineering products is written in this user manual.

Please take care that some of the products are working with a step-up converter. So, the current needed in case of undervoltage conditions must be considered higher than at the nominal voltage level. The user must choose the right fuse or other over current protection measures.

2.3 Warranty and Safety Information

	Before operating the device, read this manual thoroughly and keep it for your reference. The latest documentation for the Capture Module Serdes GMSL2/3 can be requested here:
	As a user of this device, you may only perform tasks that are described in this document and valid for the corresponding firmware version. Technica Engineering assumes no liability for any kind of damages arising from the use of or concerning the use of our devices. Full responsibility for operation rests on the side of the end user. Manufacturer warranty of Technica Engineering products only applies as long as the user ensures proper installation, configuration, and use of the device as described in this user manual. Use the device only as described in this manual. Use only in dry conditions. Do not insert any foreign object in the slots/openings of the housing. Do not apply power to a damaged device. The device is to be used only by specialists.
	Do not open the device. Otherwise, the warranty will be lost.

	This product is intended for use in automobiles or automotive-like environments. An automotive-like environment includes test setups or test benches in the office, laboratory, and workshop areas. In the test setups, the same environmental conditions apply as in-vehicle electrical systems. Technica Engineering products are not intended to be used as standard IT equipment. The test systems and products from Technica Engineering are designed as customer and application-specific test modules that are only used by specialists for the development and test facilities. When integrating the modules in a vehicle or test setup, the user must ensure appropriate ventilation or air convection for the device(s). Additionally the user (Integrator) of this product is the responsible for the external powersupply of this module, assuring the appropriate cable protection (fuse). The maximal current load value for the product is described in the user manual. Technica Engineering products must not be considered as a safety element out of context when using safety-critical systems and must be included in the safety assessment when used. The development class in a safety system must be considered with standard QM referred to as ISO26262.
	 The device can get hot. Do not cover the device due to fire danger. Do not place the device near highly flammable materials due to fire danger. Do not use the device above the specified operating temperature. The operating temperature is the ambient temperature of the installation space.
	This symbol is only valid in the European Union. If you wish to discard this product, please contact your local authorities or dealer, and ask for the correct method of disposal. Technica Engineering GmbH is registered as a manufacturer of the brand "Technica Engineering" and the device type "Small devices of Information and Telecommunications technology for exclusive use in non-private households". WEEE Reg. No. DE 20776859

3. Product description

The **Capture Module Serdes GMSL2/3** from Technica Engineering GmbH, part of our family of IVN2Eth Capture Modules, can be used to capture the traffic from up to 4 x GMSL2/3 link lines, each connecting sensors with ECUs. The traffic is captured without causing interference on the network.



Figure 3-3-1: Capture Module Serdes GMSL2/3

3.1 Functionality

The main task of the Technica Engineering Capture Modules is capturing the data of in-vehicle (IVN) communication systems via their logging network based on Ethernet. In this way, a user can set up specific logging solutions fitting the user's needs. This is even possible in selected scenarios to use a regular computer instead of a data logger.

To be able to do that, each message captured from the in-vehicle network is encapsulated in a standard transport layer frame. This requires embedding in a standard Ethernet frame according to IEEE 802.3. Data from non-Ethernet communication systems, as well as Ethernet-based systems, are transported in Ethernet frames and identified unambiguously using the protocol PLP. Ethernet frames are transported without changes to them or their headers. All messages are left intact. This ensures that analysis tools can easily access the unchanged data if support of TECMP was added or data was converted.

The main use case of the Capture Modules is to transmit the IVN data collected by the device(s) to a logging endpoint connected via Ethernet. This endpoint can be a computer, an Ethernet switch for a network connection, or a data logger module, and it is usually referred to as a *sink*. PLP specification defines the data format in which the collected data are transmitted to the configured sinks.

For more information about this protocol, please check the PLP specification document

3.2 Features

The main features of the Capture Module Serdes GMSL2/3 are summarized below:

- 4x GMSL2/3 link lines (8 ports)
- 4x SFP+ Ports for Logging data output (up to 10Gbps each)
- 1x 1000BASE-T Fast Ethernet Port for configuration
- Easy configuration via webserver and remote configuration messages
- Import and export of configurations
- Extended voltage range: 12-to-24-volt automotive battery voltage systems compatible
- Rotary switch for manual configuration of the device IP address
- Possibility to reset to default settings by the rotary switch
- Robust stainless-steel case

3.3 Technical data

Voltage requirement:	9 to 24 Volt DC (nominal 12/24 Volt DC)
Power consumption:	15 Watt (depending on connected SFP Modules)
Size:	240 x 130 x 65 mm
Weight:	1.5Kg (estimated)
International Protection:	IP 2.0
Operating temperature:	-40°Celsius to 75°Celsius

Operating temperature is the guaranteed range of temperature under which all the device features are available and working. Outside this range, the user operates the device at their own risk.

The temperature conditions where the device can be placed while not in use should not exceed 50° for the device to have consistent performance for longer periods of time.

4. Hardware

The Capture Module Serdes GMSL2/3 has been designed for multiple use cases. Therefore, several hardware versions of the Capture Module Serdes GMSL2/3 will be developed. Differences between the versions are collected in the table below:

Product version	Properties
V1.0	Initial version with DC-DC module
V1.1	Camera connector modified from HFM to Tyoc Mate-AX. Support for Reference over reverse clock forwarding.
V1.2	Updated Isolated Power connector Support for 100BaseT1 configuration port Case size reduced
V2.0	PPS signals removed
V2.1	Support for deep sleep Support for PoC injection
V2.2	Production improvements
V3.0	Support for PoC injection based on external voltage source Support for PoC injection based on configurable internal voltage source

Please check the label at the bottom of your device to know what version of the hardware you have and pay attention to its characteristics and limitations for proper use.

4.1 Connectors and controls

The Capture Module Serdes GMSL2/3 is equipped with several connectors and control elements that are distributed between the front and back sides of the device.

On the front side (Figure 4-1) of the Capture Module Serdes GMSL2/3, we can find:

- 1x Isolated power connector including
  - 1x Wake up lines
- 1x Rotary switch
- 1x Non-isolator power connector including
  - 1x 100BASE-T1 port
  - 1x Wake up lines
  - 1x External PoC voltage source (introduced in Product version 3.0)
- 1x RJ-45 Ethernet connector (for configuration and synchronization)
- 1x RJ-45 Ethernet connector (for configuration and synchronization)

- Several LEDs
- 4x SFP+ slots

The back side (Figure 4-2) of the device is reserved for the interfaces used to capture traffic from the in-vehicle network (IVN). In the case of a Capture Module Serdes GMSL2/3, this means:

- 4x GMSL2/3 link lines (= 8x GMSL2/3 ports)
- 1x Service port

Here user can see an overview of all the HW interfaces of the Capture Module Serdes GMSL2/3.

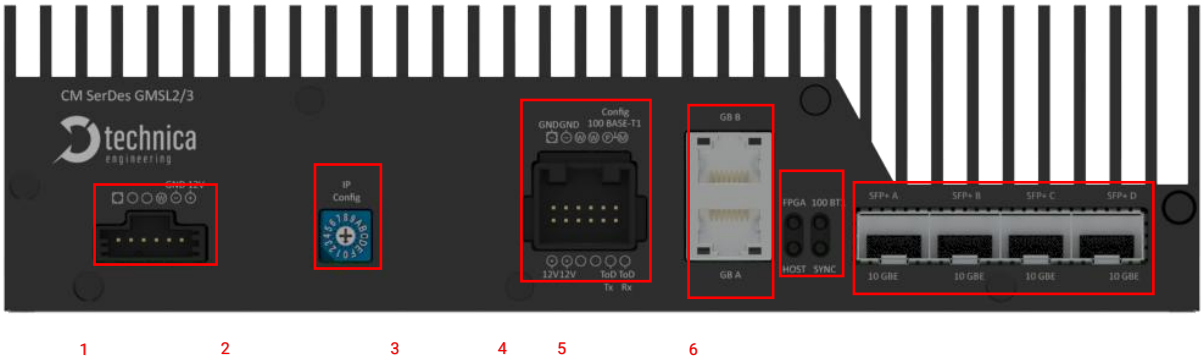


Figure 4-1: Front side of Capture Module Serdes GMSL2/3

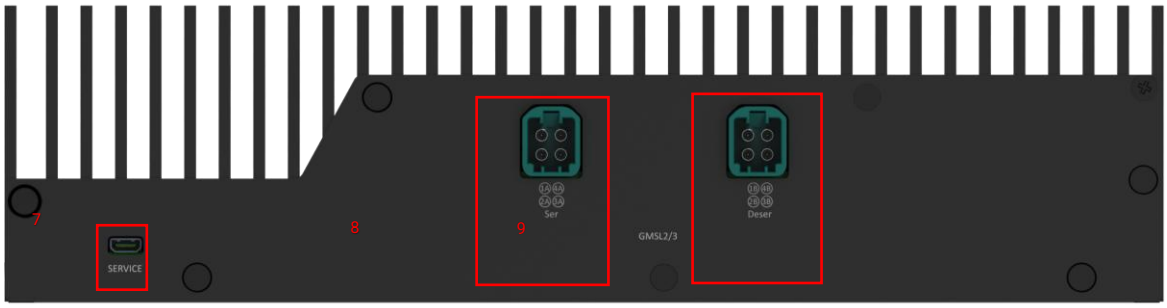


Figure 4-2 Backside of Capture Module Serdes GMSL2/3

4.1.1 Isolated power connector (1)

A black MQS connector on the front side (6 pins, one row) is used to supply the power through a galvanic isolated DC/DC module to the Capture Module Serdes GMSL2/3.

- For HW versions 1.0 and 1.1 the power supply for the device is supplied by pin 1 and 2 (12 Volt) and pin 3 and 4 (Ground).
- For HW versions 1.2 and above the power supply for the device is supplied by pin 6 (12 Volt) and pin 5 (Ground).

- Requirements for the device itself: 12 Volt DC up to 2.5 amperes.

Next to the pins for the power supply, the pin corresponding to the galvanic isolated wake-up line can be found, identified with the letter “W” in the device housing. This pin are be able to act as a digital input.

The connector Tyco Electronics Micro Quad Lock System (MQS) is used, as described in the table below:

Name	Picture	Part Number
TE Connectivity MQS BU-GEH 6P		1-969508-2 Distributor: mouser.de
TE Connectivity MQS crimp contact		928999-1 Distributors: Mouser.de; Digikey.de

Table 2: Parts of Tyco MQS Connector

Pin	Function
1	Battery +12 Volt Input
2	Battery +12 Volt Input
3	Power Ground
4	Power Ground
5	Isolated Wake-up line
6	n.c.

Table 3: Pinning of black MQS connector for HW version 1.0 and 1.1

Pin	Function
1	n.c.
2	n.c.
3	n.c.
4	Isolated Wake-up line
5	Power Ground
6	Battery +12 Volt Input

Table 4: Pinning of black MQS connector

4.1.2 Rotary Switch (2)

A rotary switch is located on the left-most part of the front side of the Capture Module Serdes GMSL2/3. This element allows the configuration of the IP address of the device associated with the GB ports (i.e., the 1000BASE-T ports, with RJ45 connector). If any of them is set as configuration ports. This is the IP address that will be used to access the web server for device configuration.

With the rotary switch, it is possible to select an IP address to access the device. This is the so-called HW IP address, a shortcut for *HW-determined IP addresses*.

For each hardware variant of Capture Modules, a different IP address range has been defined. In the case of a Capture Module Serdes GMSL2/3, the following IP addresses will be set:

Rotary switch on 0 -> IP: 10.104.3.192
Rotary switch on 1 -> IP: 10.104.3.193
Rotary switch on 2 -> IP: 10.104.3.194
...
Rotary switch on E -> IP: 10.104.3.206
Rotary switch on F -> IP: 10.104.3.207 / Reset

Rotary switch in position 0 to E

If the rotary switch is set to a position from *0* to E, the IP address of the device is not configurable via software and can only be changed using the rotary switch. The IP address will be configured according to the table above.

When changing the rotary position to any of these positions, the corresponding IP address is set after a software reset. The IP address currently in use is shown in the *System Information* tab on the device website.

Rotary switch in position F

Position F offers the user the possibility to perform a factory reset of the device. To do this, the rotary switch must be changed to position F, followed by a power reset performed by the user. The **complete configuration** will be lost and set to default, including the IP addresses:

- Controller IP address: 192.168.0.49
- HW IP address: 10.104.3.207

If the device is blocked and all LEDs are lit (not blinking), the device must be sent to Technica Engineering. To avoid problems please make sure that the Host LED is blinking in its normal operation mode and that the power supply is stable before the reset.

4.1.3 Non-isolator power connector (3)

A black MQS connector on the front side (12 pins, two rows) is used to supply the power without isolation to the Capture Module Serdes GMSL2/3.

- The power supply for the device is supplied by pin 7 and 8 (12 Volt) and pin 1 and 2 (Ground)
- Requirements for the device itself: 12 Volt DC up to 2.5 amperes.

Next to the pins for the power supply, the pins corresponding to two independent wake-up lines can be found, identified with the letter “W” in the device housing. Both of them are be able to act:

- As a digital input.
- As a digital output.

Apart from that, in this connector, we can find two pins corresponding to a 100BASE-T1 port. Through this 100BASE-T1 port, it will be possible to get access to the web server used for device configuration using the default IP address (192.168.0.49, if not changed).

A converter between 100BASE-T1 (automotive Ethernet) and 1000BASE-T (Gigabit Ethernet, RJ45) is required to connect directly to the device from a computer.

Finally, this connector has 2 pins dedicated to receiving the Time of Day (for synchronization purposes) from an external device through a serial interface. **The function associated with these pins is currently not supported by the software.**

The connector Tyco Electronics Micro Quad Lock System (MQS) is used, as described in the table below:

Name	Picture	Part Number
Tyco, MQS Connector 12 Pins Black		1-1534096-1
Tyco 12 pins MQS Socket		1534100-1
Tyco crimp contact		928999-1

Table 4-5: Parts of black MQS connector

The official Tyco tool can be used for these crimp contacts. A less expensive variant is the crimp tool for “PSK” contacts.

Pin	Function	Pin	Function
1	Power Ground	7	Battery +12 Volt Input
2	Power Ground	8	Battery +12 Volt Input
3	WakeUP 1	9	n.c.
4	WakeUP 2	10	n.c.
5	100BASE-T1 Plus	11	ToD TX
6	100BASE-T1 Minus	12	ToD RX

Table 4-6: Pinning of black MQS connector for product versions starting lower than 3.0

Pin	Function	Pin	Function
1	Power Ground	7	Battery +12 Volt Input
2	Power Ground	8	Battery +12 Volt Input
3	WakeUP 1	9	External voltage source for PoC Injection
4	WakeUP 2	10	External voltage source for PoC Injection
5	100BASE-T1 Plus	11	ToD TX
6	100BASE-T1 Minus	12	ToD RX

Table 4-7: Pinning of black MQS connector for product versions starting from 3.0

4.1.4 1000BASE-T (RJ45) connectors (4)

There are two RJ45 Standard Ethernet connectors for Gigabit Ethernet on the front side, labeled GB-A to GB-B, that can be used for the configuration of the device.

On future SW releases, these ports can be used as well for time synchronization between different IVN2Eth Capture Modules of Technica Engineering GmbH.

4.1.5 LEDs (5)

The Capture Module Serdes GMSL2/3 has 4 Status LEDs on the front side of the case.

LED	Description
Host LED	Orange: Blink slowly (once/second) during normal operation of the device Blink fast (flicker) suggesting the device is in bootloader mode

FPGA LED	Green: shows the state of the FPGA, in charge of the processing of the captured data. It blinks with a frequency of approx. 0.5 sec during normal operation. If it is not blinking or blinking faster, FPGA is damaged and must be repaired.
100 BT1	Green: will show the linkup state of the 100BASE-T1 port connected directly to the microcontroller.
SYNC	Orange: will show the state of the device synchronization with a distributed clock (through PTP)

Table 8 Details for Status LEDs

4.1.6 SFP+ (6)

Four SFP+ slots facilitate a 10Gbps connection for logging the captured traffic. The Capture Module Serdes GMSL2/3 only supports 10Gbps SFP+ modules (with 10GBase-R interface).

The following modules have been tested and can be used with the Capture Module Serdes GMSL2/3:

- 10Gtek 10GBASE-T SFP+ Transceiver, RJ-45 SFP+ CAT.6a up to 30 meters.
- FS 10GBASE-T SFP+ Transceiver, RJ-45 SFP+ CAT.6a up to 30 meters.
- FINISAR FTLX8573D3BTL Optical module
- EDGE 10G-SFP-300I_MS Optical module

Webpage access (for configuration) is via SFP+ port not possible

4.1.7 Service (7)

The Capture Module Serdes GMSL2/3 is an interface for programming and debugging, of type micro-HDMI, whose use is reserved only for development.

4.1.8 GMSL2/3 Serializer Port (8)

The Capture Module Serdes GMSL2/3 is provided with an HFM connection with 4 mini-coax ports connected to GMSL2/3 serializers. These ports are intended to be connected to vehicle ECUs. They use MAX96793 serializers.

4.1.9 GMSL2/3 Deserializer Port (9)

The Capture Module Serdes GMSL2/3 is provided with an HFM connection with 4 mini-coax ports connected to GMSL2/3 deserializes. These ports are intended to be connected to vehicle sensors. They use MAX96792A deserializers.

5. Configuration

The Capture Module Serdes GMSL2/3 is a highly configurable device, offering the user great flexibility to include it in many different scenarios.

Out-of-the-box, access to the web server is possible by connecting the device via the 1000BASE-T GB-A port (RJ45) to a computer. The default IP address for the device is 10.104.3.192 (with rotary switch on position “0”). The recommended subnet mask is 255.255.255.0.

Once connected to the device (through the 1000BASE-T GB-A port):

1. Please go to the network adapter settings and adjust the IP in the same range as Capture Module Serdes GMSL2/3. For example, 10.104.3.20 with subnet mask 255.255.255.0.
2. Enter the IP address in your standard web browser.

Google Chrome is the recommended browser.

5.1 Home Screen

The following website should appear. The user can select the corresponding ports for further configuration

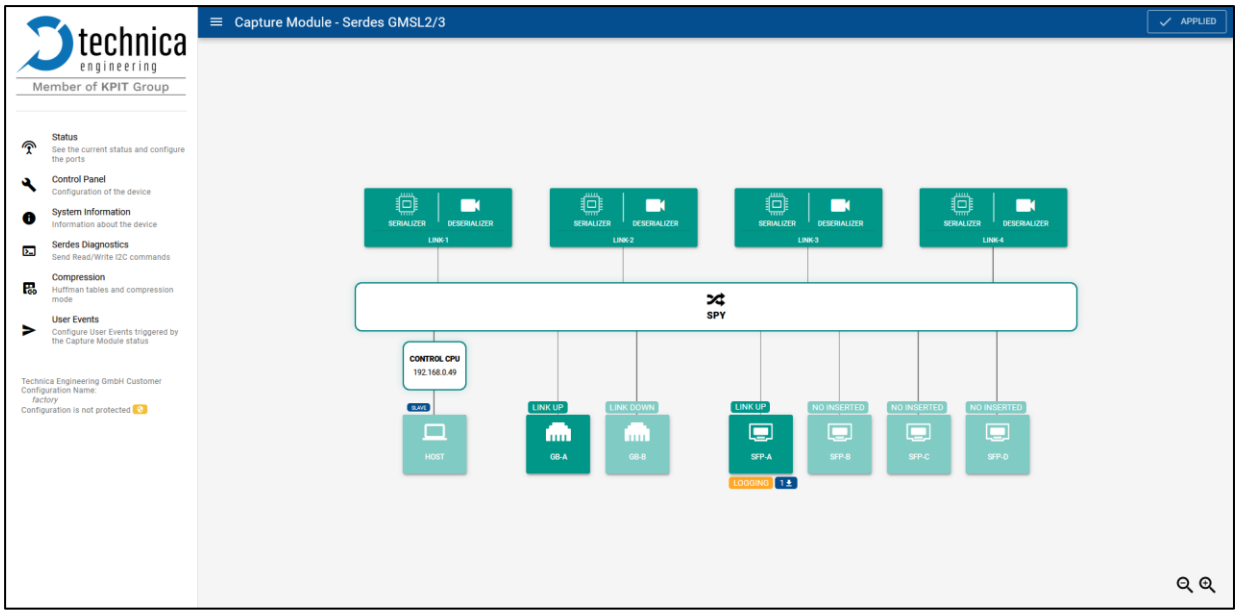


Figure 5-1: Home Screen Capture Module Serdes GMSL2/3

This home screen includes links to the different sections available, which allow the user to get information about the Capture Module Serdes GMSL2/3 and its configuration.

Please select one of the tabs for further configuration. Each tab is described in detail in the next sections of this document.

Any change in the configuration will lead to the following hint appearing at the top of the current web page:





This option sends data from the webpage to the memory of the device. There is no need to save every time this hint appears. All the desired changes can be done at once and the configuration can be saved at the end. To do that, please click "SAVE CONFIGURATION". Please note that the configuration will not be applied unless the device is restarted.



By clicking on this option, the user is prompted to introduce an optional name for the configuration file and to accept the changes. By selecting "OK" the configuration is saved similar to the option "SAVE CONFIGURATION". It also restarts the device, applying the saved configuration.

Note that without " SAVE & APPLY CONFIGURATION", the configuration of the device will not be fully updated. To be sure that all changes have been applied, a restart of the device is necessary.

5.2 Status tab

The following options can be explored in the Capture Module Serdes GMSL2/3 configuration tab.

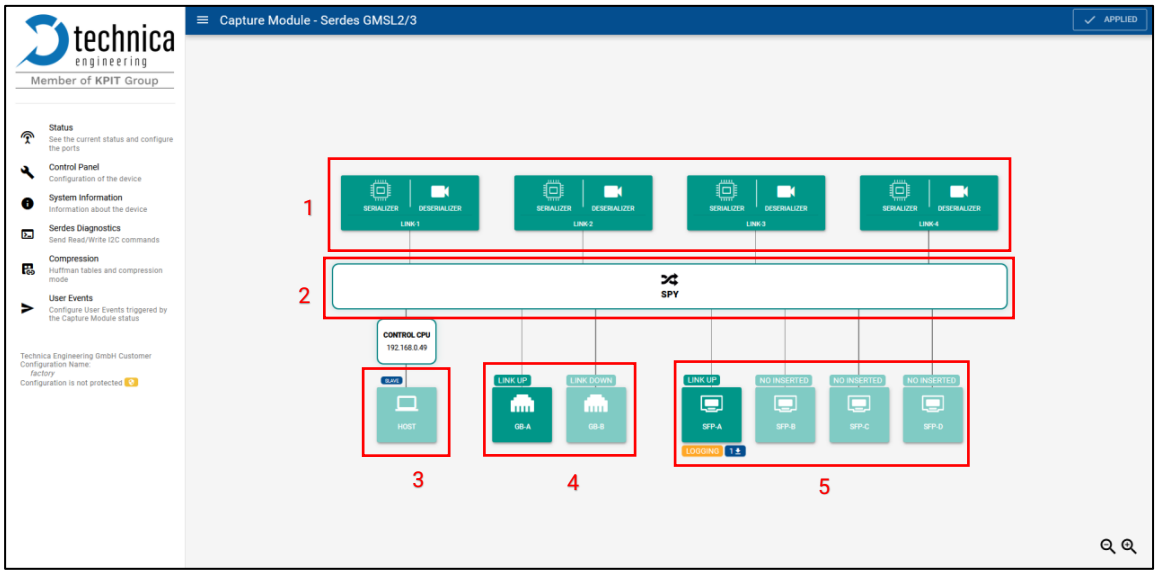


Figure 5-2 Status Configuration Tab

5.2.1 GMSL2/3 Links (1)

The Capture Module Serdes GMSL2/3 is provided with four-link lines. Each of these lines is composed of one GMSL2/3 serializer port and one GMSL2/3 deserializer port where a sensor and an ECU can be connected to capture the communication between them.

Each of the GMSL2/3 links requires a basic configuration to determine how exactly the Capture Module will handle the traffic captured from the DUT connected to it if any.

The user can find in the switch status tab a graphical view of the device, with the four GMSL2/3 links. Clicking on one of the links, a configuration page that looks as follows will be opened:

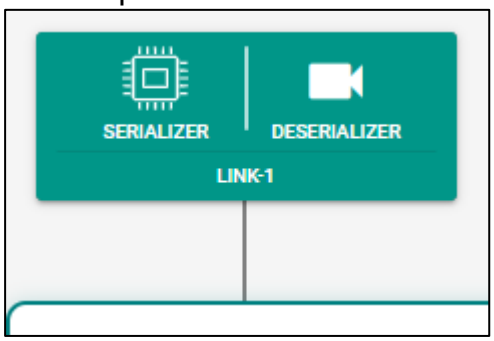


Figure 5-3 GMSL2/3 Link Visualization

Depending on the status of the GMSL links initialization sequence, the color of the GMSL link is shown as follows:

- Dark Green: Initialization sequence completed successfully

- Light Green: Initialization sequence not completed
- Grey: An error has been detected during the GMSL initialization sequence

5.2.1.1 Logging Configuration

Each CM Serdes GMSL2/3 link can be configured independently for logging all the received data (video, I2C and GPIO). The logging functionality of the CM Serdes GMSL2/3 can be configured through the following options:

- I2C 1 Bus ID: Busspec ID used to log packets from the control I2C bus.
- I2C 2 Bus ID: Busspec ID used to log packets from the pass-through I2C bus.
- GPIO Bus ID: Busspec ID used to log packets from the GPIO events.
- Output to: SFP+ port to send all the captured packets.

Keep awake reason channel

The GMSL ports can be taken into consideration to decide whether the CM Serdes GMSL2/3 can enter the sleep mode (if the device is configured for that). For that, the port will be monitored to detect incoming traffic.



Keep awake reason channel selected: Port is monitored. If traffic is detected on this port, the device must stay in full operation and cannot enter sleep mode.



Keep awake reason channel not selected: Port is not monitored. The device can enter sleep mode even if there is traffic on this port

Video Logging General Configuration

The configuration of the Video Logging can be modified though the following parameters.

- Payload Size: The user can select the payload size of the packets, from the following options: 1000 bytes, 2000 bytes, 4000 bytes, 6000 bytes.
- Pixel Arrangement: The user can select the arrangement of data within the packets, from the following options:
    - 0-padded byte aligned: Uncompressed RAW data with 0-padding: RAW12/20/… values are 0-padded and byte-aligned, i.e. 1x RAW12 = 2x UINT8, 1x RAW20 = 3x UINT8, …
    - 0-padded word aligned: Uncompressed RAW data with 0-padding, aligned to word boundaries: RAW12/20/… values are 0-padded and word-aligned, i.e. 1x RAW12 = 1x UINT16, 1x RAW20 = 2x UINT16, …
    - Densely Packed: Uncompressed RAW data, densely packed: RAW12/20/… are densely packed using a “gearbox”, i.e. 2x RAW12 = 3x UINT8
- FPGA Video Path Enabled flag: Enables video capture channels.
- Compression: The user can enable Lossless JPEG compression for each GMSL2/3 Link.

- FPGA Video Path Compression flag: Video path to be compressed.
- Compression mode: The user can select frame-based or line-based compression.
- Adaptative compression: Enables a more efficient and dynamically updated compression model by adapting to the data as it is processed.

Compression can only be enabled for up to a total of 4 video paths across all the GMSL links.

Lossless JPEG compression requires that the received video always have the same resolution. If lesser lines are received than expected, the outgoing compressed stream will not be correct.

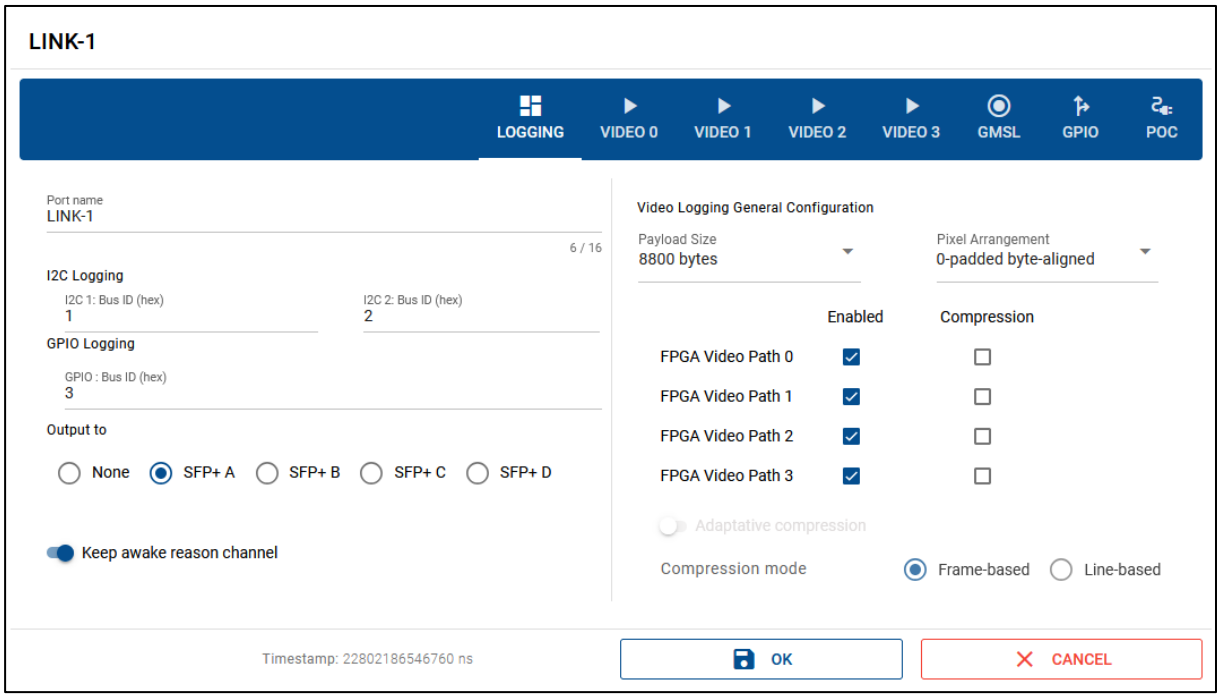


Figure 5-4 Logging configuration

5.2.1.2 Video Configuration

For each Video Configuration tab (corresponding to each FPGA video path), an independent Virtual Channel and BusSpecID can be configured.

These settings influence the metadata packet generated by the CM SerDes GMSL2/3 for each received frame, which includes basic information such as image height and width. Some of the fields of this packet can be configured through the webpage as follows:

- **Imager Type**: The user can define an arbitrary image type (in hex format) for image sensor identification.
- **RAW Color Scheme**: The user can select the color scheme (pixel arrangement) of the sensor.
- Frame Count:
    - **Frame Count extractor**: Enable the decoding of the frame count from the image fames. Bytes appear in the ethernet packet in the same order as they were received on the MIPI interface
    - For each frame counter byte, the user can configure the following parameters:
        - **Image line**: Image line where the frame count is located. The first image line of the frame is **1.**
        - **Offset:** Offset from the first image line byte where the frame count is located. The first byte of the image line (offset) is **0.**
        - **Enable flag :** If enable flag is disabled, the CM Serdes GMSL2/3 will set the associated frame counter byte to 0x00 (default value).
- **MIPI Datatype Overwrite:** The user can select a MIPI datatype to overwrite the datatype of captured MIPI long packets per virtual channel. This setting only applies to MIPI long packets with a video datatype.

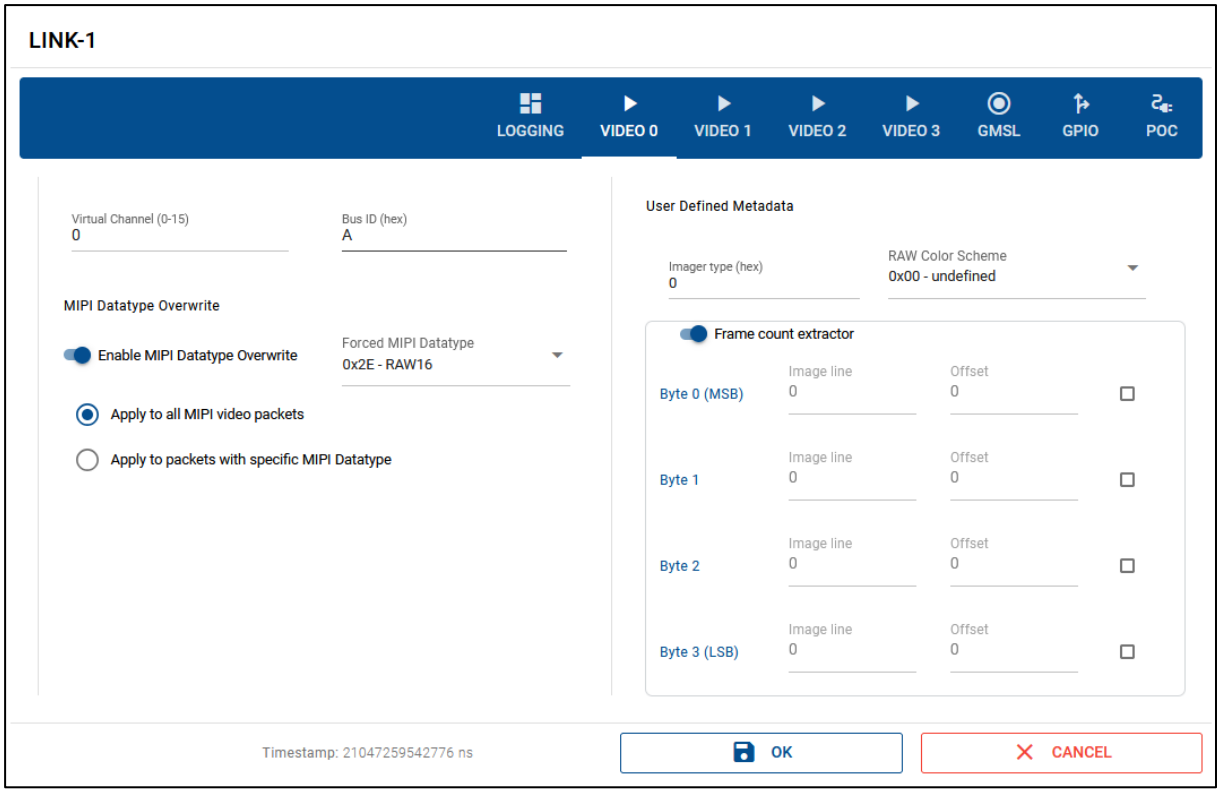


Figure 5-5 FPGA Video Path configuration

5.2.1.3 GMSL Configuration

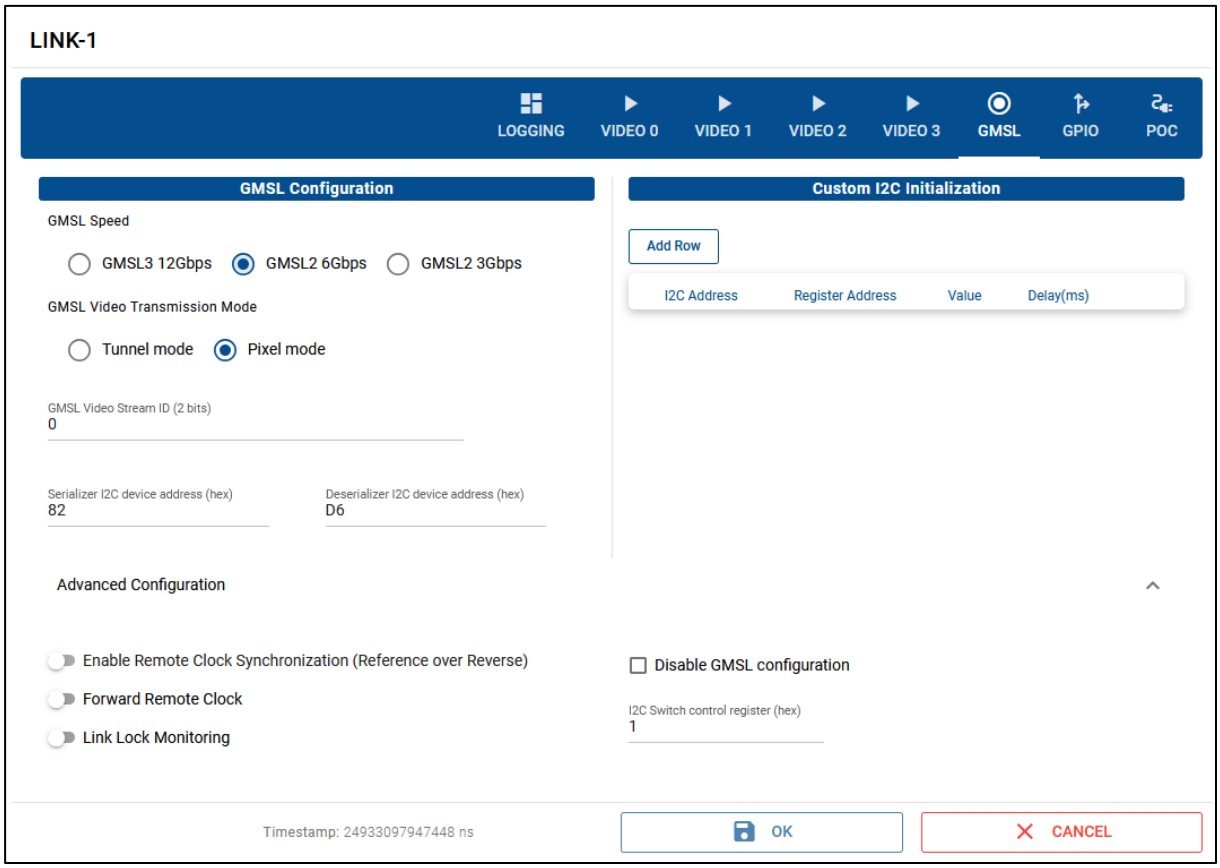


Figure 5-6 GMSL2/3 Link configuration

GMSL configuration

Each GMSL2/3 link can be configured with the following basic options:

- GMSL Speed: Speed can be set to 12Gbps (GMSL3), 6 Gbps, or 3 Gbps(GMSL2)
- GMSL Video Transmission Mode: GMSL video transmission mode can be set to pixel or tunneling
- GMSL Video Stream ID: definition of GMSL stream ID (used for video forwarding and replication)
- Serializer I2C device address: The serializer I2C device address can be modified to avoid having devices with duplicated I2C address on the same I2C bus.
- Deserializer I2C device address: The serializer I2C device address can be modified to avoid having devices with duplicated I2C address on the same I2C bus.

These settings define the basic GMSL2/3 link initialization sequence.

Custom I2C Initialization

The basic initialization sequence can be extended by I2C operations to be performed on the initialization of the device. Write operation to any device can be configured and also a delay between operations can be added.

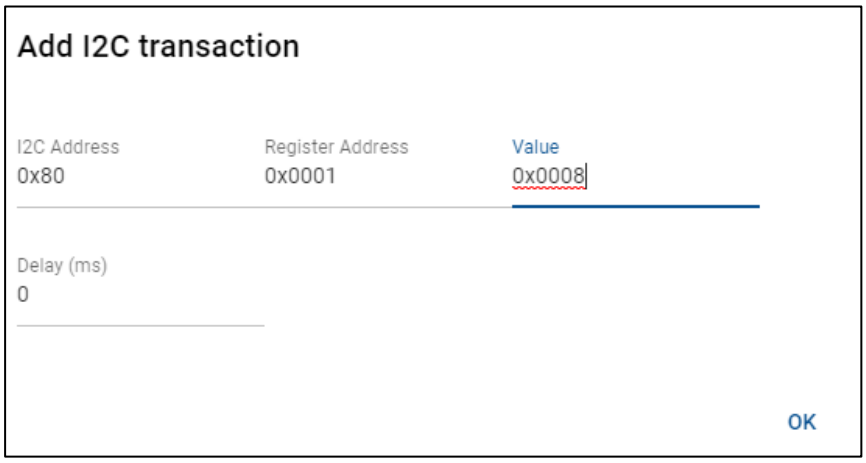


Figure 5-7 I2C transaction configuration

Advanced Configuration dropdown menu

- Enable Remote Clock Synchronization: When this setting is enabled, the The CM Serdes GMSL2/3 is configured to recover the clock from the ECU (using the Reference over Reverse feature on the serializer).
- Forward Remote Clock: When this setting is enabled, the CM Serdes GMSL2/3 is configured to forward the clock recovered from the ECU (using the previous option) and forwards it to the camera (by forwarding the recovered clock from the CM Serdes GMSL2/3 serializer to the deserializer). When this setting is disabled, a fixed 25MHz clock is selected to be forwarded to the camera.
- Link Lock Monitoring: Monitors the GMSL link status and in the event of a link-down detection, triggers a reset of the internal serializer and deserializer.
- Disable GMSL configuration flag: Set this flag to disable the system's GSML configuration.
- I2C switch control register (hex): The data Capture Module Serdes GMSL2/3 includes an analog based I2C switch which can be controlled through the webpage to allow different I2C network topologies. This setting can be configured from the website.

5.2.1.4 GPIO Configuration

Each GMSL devices has MFP pins (multifunction pins) that are used to transmit control signals and generic GPIO between the local and remote GMSL devices through the GMSL reverse link (side channel operations). An application of the MFP usage is to transmit frame synchronization signals from an ECU to all the cameras to assure that all images are captured at the same moment.

The CM Serdes GMSL2/3 allows the transmission of GPIOs between the serializer ports and deserializer ports while also logging all the events to the datalogger. To configure the transmission and logging of GPIOs, the CM Serdes GMSL2/3 allows the following configurations:

- GPIO Mapping: Configuration of the serdes devices MFP pins for transmission or reception
- GPIO Bridging: Connection of MFP pins within the CM Serdes GMSL2/3
- GPIO Logging: Logging of the serdes devices MFP pins to ethernet frames

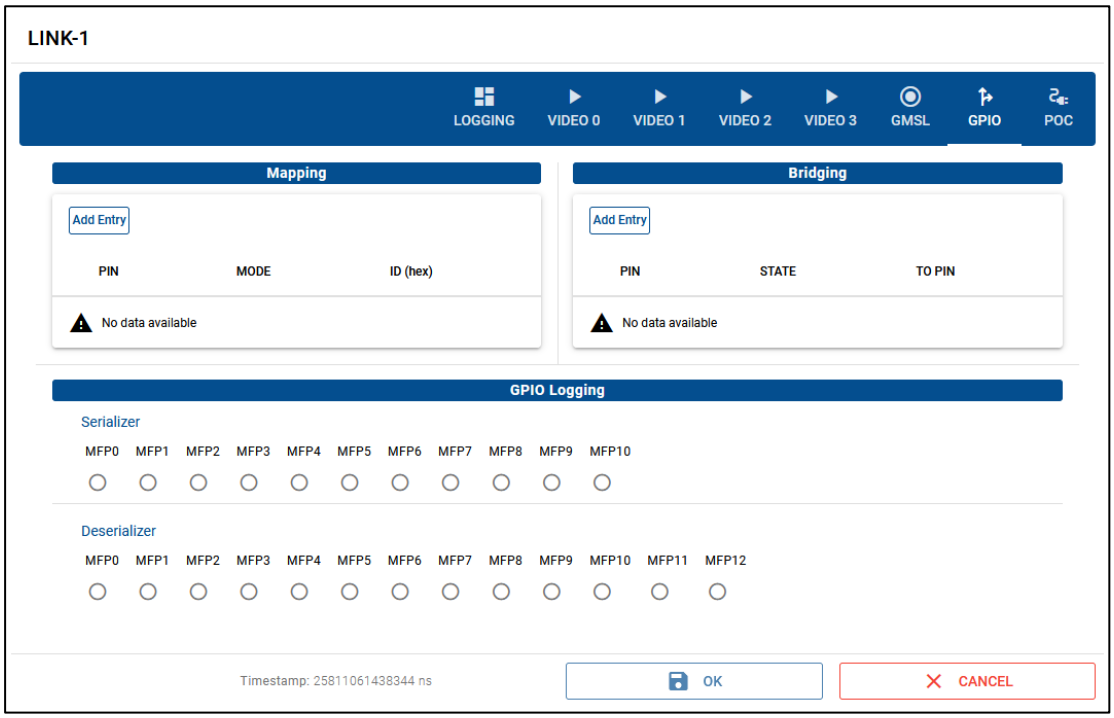


Figure 5-8 GMSL2/3 Link configuration

GPIO mapping configuration

All serializer and deserializer MFP pins configured to be transmitted over the GMSL reverse link. The pins can be configured on the following modes:

- TX → The CM Serdes GMSL2/3 configures the specified pin for transmitting its status over the GMSL reverse link using the transmission ID specified on the ID field.
- RX → The CM Serdes GMSL2/3 configures the specified pin for receiving its status over the GMSL reverse link using the reception ID specified on the ID field.

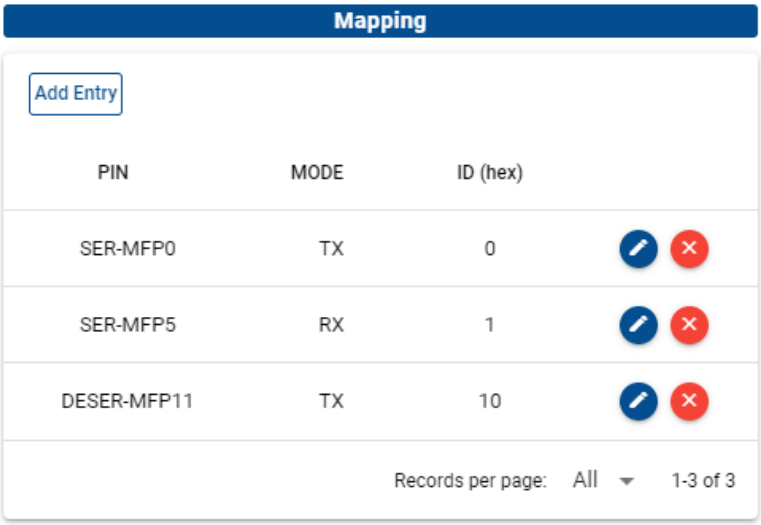


Figure 5-9 GPIO mapping configuration

GPIO switching configuration

All serializer and deserializer MFP pins can be switched by configuring the internal MAX10 connected between the serializer and deserializer. For each serializer and deserializer MFP the following configurations are available on the webpage:

- Low → The CM Serdes GMSL2/3 forces this MFP to 0
- High → The CM Serdes GMSL2/3 forces this MFP to 1
- High-Z → The CM Serdes GMSL2/3 leaves the pin in High impedance. This is the default state for all the MFP pins
- Bridge → The CM Serdes GMSL2/3 forwards the MFP from the serializer to the deserializer or from the deserializer to the serializer.

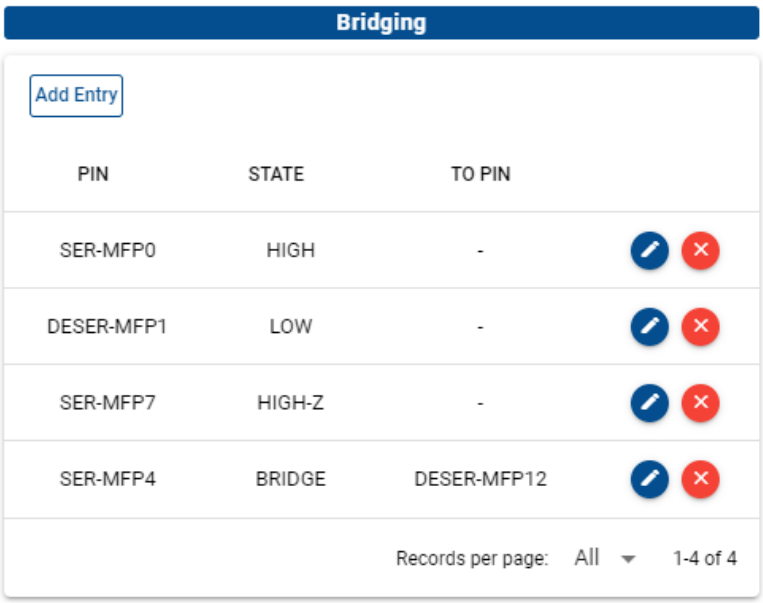


Figure 5-10 GPIO switch configuration

GPIO Logging configuration

All serializer and deserializer MFP pins can be configured to be logged with PLP frames or not. By default, logging from all MFP pins is disabled.

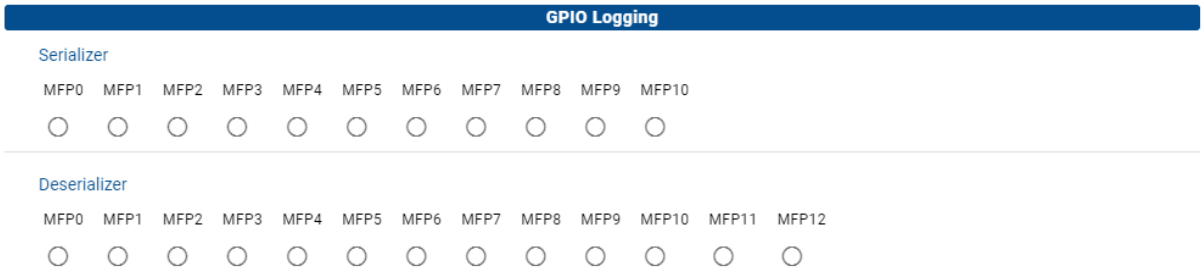


Figure 5-11 GPIO Logging configuration

5.2.1.5 POC Configuration

The CM Serdes GMSL2/3 can be configured to inject 12V to the camera (instead of bypassing) with a maximum injected current of 600mA. When this setting is enabled, the CM Serdes will inject 12V and will monitor the current drawn by the camera and several error conditions. Current and error conditions are shown on the webpage:

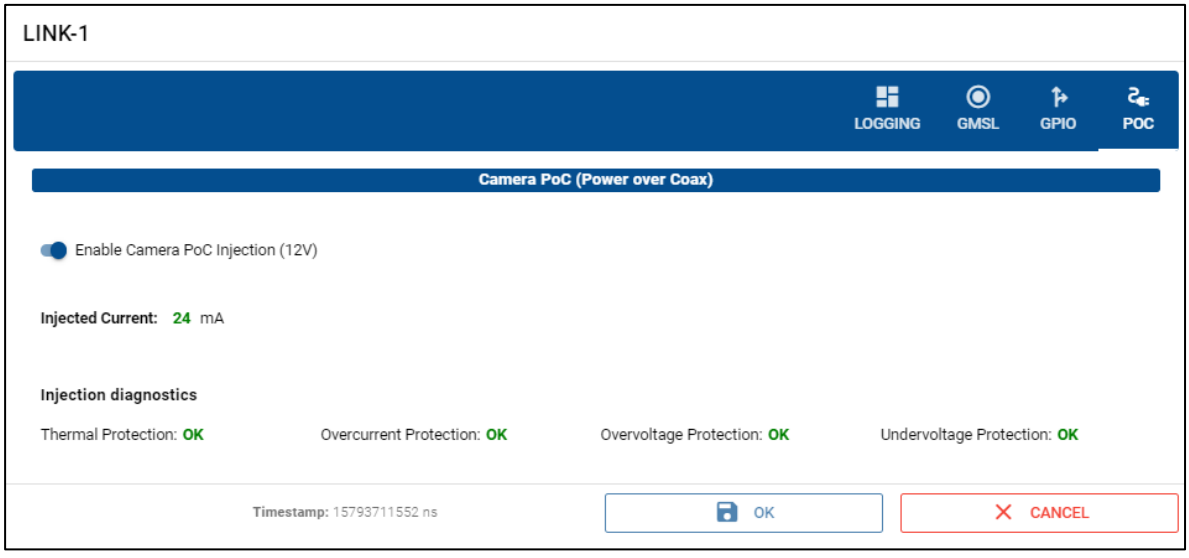


Figure 5-12 GMSL2/3 Link configuration

5.2.2 SPY (2)

The Spy multiplexer is the part in charge of the logging activities. By clicking on the Spy element in the graphical overview, the user has access to the general logging configuration.

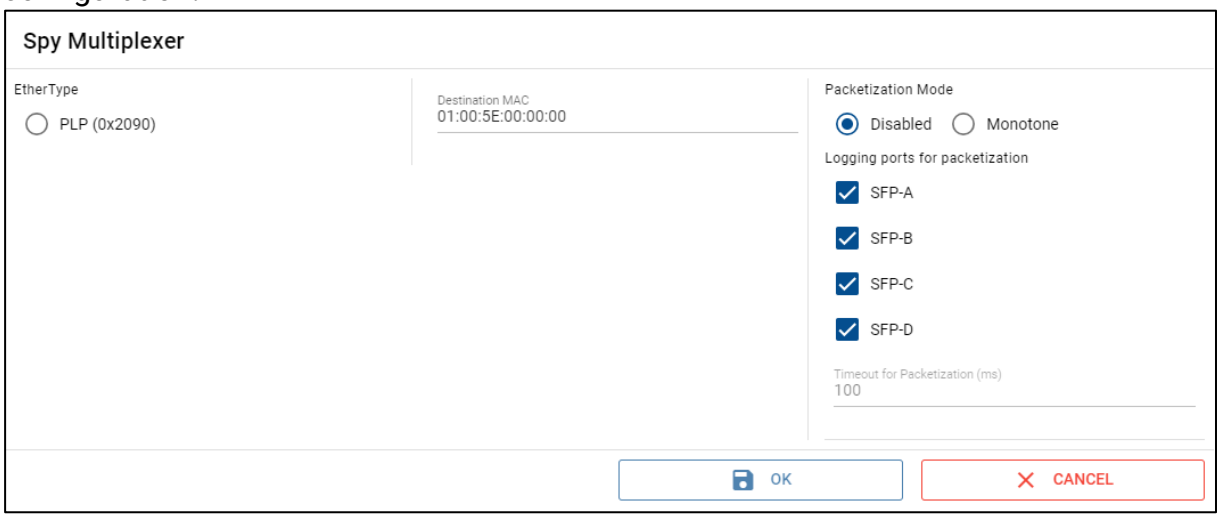


Figure 5-13 Spy multiplexer

5.2.2.1 Destination MAC

For encapsulation, layer 2 (data link layer of the OSI model) is used. The encapsulated Ethernet frames are sent to a multicast MAC address. Default value is 01:00:5E:00:00:00.

5.2.2.2 Packetization mode

Packetization is a feature that allows encapsulation of more than one captured frame into one PLP frame. This helps to increase the performance of the transmission of the captured data to a data logger or a computer, especially in case the traffic is composed mainly of small frames.

- Disabled: Each incoming Ethernet frame will be encapsulated in a separate PLP frame.
- Monotone: Several Ethernet frames will be combined in a PLP frame, guaranteeing that the timestamp of the PLP frames will increase always monotonously for a given interface ID.
    - For each interface ID, all the captured frames will be sent out encapsulated in PLP frames in the same sequence as they were captured.

5.2.2.3 Logging Ports for packetization

CM Serdes GMSL2/3 can handle packetization in all the output ports at the same time. With this option, packetization can be enabled separately for each output port configured as a logging port.

5.2.2.4 Timeout for packetization (ms)

The amount of Ethernet frames included in a packetized PLP frame is defined by two conditions:

- Length: Maximum PLP payload length of 700 bytes.
- Timeout for packetization: Maximum time the device will wait to fill in a PLP frame with as many captured frames as possible.
    - Allowed values for the timeout are between 100 ms (default) and 65535 ms.

If one of the limitations is reached, the PLP frame is sent out through the logging port.

5.2.3 HOST port (3)

This is the 100BASE-T1 port located in the MQS connector on the front side of the Capture Module Serdes GMSL2/3. It is directly connected to the microcontroller that hosts the webserver used for device configuration (hence the name).

Used as a fallback solution for flashing in case of the usual procedure via GB port is not possible.

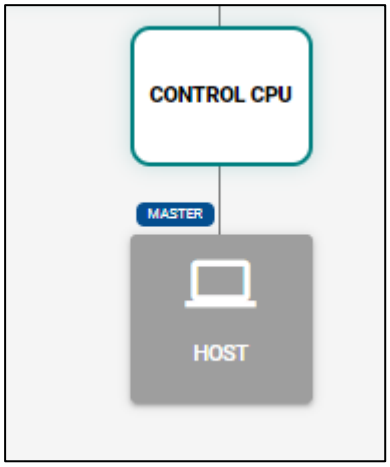


Figure 5-14 Control CPU & HOST port

5.2.4 GB ports (4)

The Capture Module Serdes GMSL2/3 is equipped with two GB ports. These ports implement the 1000BASE-T standard and can be used as the main interface to interact with the device and connect it to the logging network. They are used for configuration and time synchronization.

The user can find in the *status tab* a graphical view of the device where the GB ports are located on the front of the device, analog to their location in the physical device.

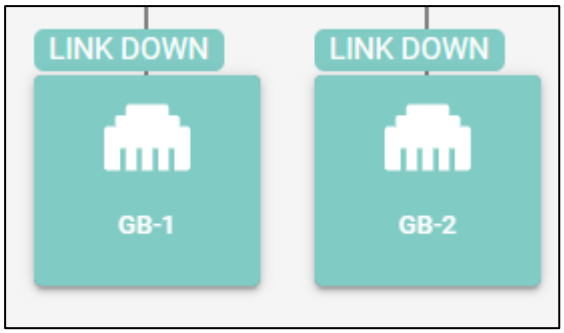


Figure 5-15 GB ports

5.2.4.1 Configuration port

This option lets the user select this port to access the web server for device configuration.

Configuration port selected: The device can be configured by connecting a computer to this port and accessing from it the configuration website.

Configuration port not selected: Configuration website is not reachable from this port, thus not possible to use this port for configuration.

By default, the GB-A port is set as a configuration port. In case no port is selected as a configuration port, the GB-A port will be used as well.

Only one port can be selected as a configuration port

5.2.4.2 802.1AS Mode

The Capture Module Serdes GMSL2/3 offers the time master functionality for IEEE802.1AS. If decided to use the device as a time master in the logging network, then one and only one port can work as a time master.

In case IEEE 802.1AS is selected as the time synchronization protocol in the system where the Capture Module Serdes GMSL2/3 will be used, then the user is requested to specify explicitly which of the ports will act as time master (if any) using this option to avoid possible misconfigurations.

Time synchronization between the Capture Module Serdes GMSL2/3 and the logging endpoint and/or other Capture Modules is possible by enabling this option in the GB port(s) where synchronization with the connected devices is required.

A GB port can be configured as follows for time synchronization:

- Disabled: No time synchronization is performed through this port.
- Slave: The Capture Module Serdes GMSL2/3 acts as an intermediate or end device that gets its time from another device, connected directly to it, which is acting as bridge master.
  - If there is already one port defined as SLAVE in the device, no other port can be configured as a SLAVE, nor as 802.1AS MASTER.
  - Once Slave has been selected the user has a choice to configure the port as:
    - gPTP Automotive slave: 802.1AS slave
    - PTPv2 Slave: IEEE 1588-2008 slave
- Master: The Capture Module Serdes GMSL2/3 acts as intermediate device that gets its time from the grandmaster (directly or through other devices) and acts as a time master for the next device in the chain. It forwards the corrected time information to all the other attached time-aware systems. Sync / Follow up messages start to be sent only when a slave device is detected.
- Grandmaster: This port is generating the master clock used as a reference for the logging system. The device will be providing the time base for all directly attached time-aware systems.

Note that it is not possible to have two ports configured as grandmaster on the same Capture Module. If selected, nothing can be configured on the other GB ports.

If no time synchronization is required, this setting should be set as "DISABLED". Otherwise, it is important to distinguish whether the synchronization will be done using gPTP or PTPv2 in the logging network.

By default, time synchronization is disabled on all GB ports.

5.2.4.3 Keep awake reason channel

The GB ports can be taken into consideration to decide whether the CM Serdes GMSL2/3 can enter the sleep mode (if the device is configured for that). For that, the port will be monitored to detect incoming traffic.

Keep awake reason channel selected: Port is monitored. If traffic is detected on this port, the device must stay in full operation and cannot enter sleep mode.

Keep awake reason channel not selected: Port is not monitored. The device can enter sleep mode even if there is traffic on this port

5.2.5 SFP+ ports (5)

The Capture Module Serdes GMSL2/3 is equipped with four SFP+ ports. These ports can be configured as 10G with 10Gbps SFP+ modules and can connect to the logging network.

The function of these ports is for logging the captured traffic.

The user can find in the switch status tab a graphical view of the device where the 10Gbps port is located on the right side of the device, analog to its location in the physical device. Clicking on the ports, a configuration page that looks as follows will be opened:

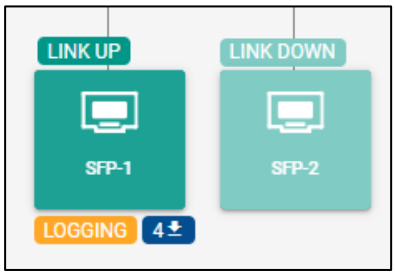


Figure 5-16 SFP+ Port Visualization

5.2.5.1 Port configuration

The user can click on the desired port for configuration. The name of the logging port can be modified. Speed mode for the SFP+ module can also be selected (the **current SW implementation only supports 10GBASE-R**). This tab also shows the logging configuration for GMSL2/3ports.

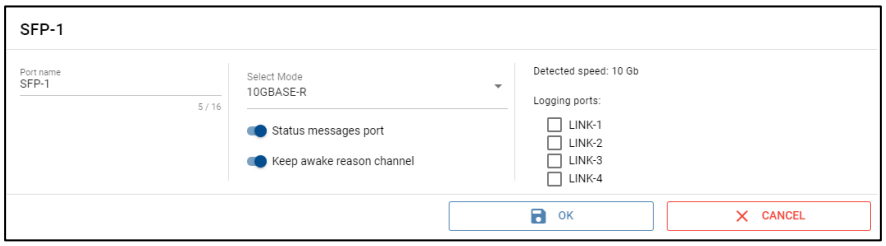


Figure 5-17 Configuration menu for SFP+ 1 Port

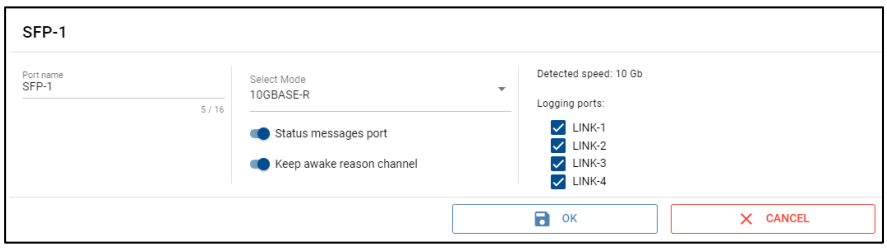


Figure 5-18 Configuration menu for SFP+ 1 Port with logging configuration from all GMSL 2/3 links

5.2.5.2 Keep awake reason channel

The SFP+ ports can be taken into consideration to decide whether the CM Serdes GMSL2/3 can enter the sleep mode (if the device is configured for that). For that, the port will be monitored to detect incoming traffic.



Keep awake reason channel selected: Port is monitored. If traffic is detected on this port, the device must stay in full operation and cannot enter sleep mode.



Keep awake reason channel not selected: Port is not monitored. The device can enter sleep mode even if there is traffic on this port

5.2.5.3 Status messages port

The CM Serdes GMSL2/3can send debug messages with information about the status of the device itself, its ports, and configuration. This can be configured by the user.



Status messages port selected: The status messages of the Capture Module are transmitted on this port



Status messages port not selected: The status messages of the Capture Module are not transmitted on this port

By default, all SFP+ ports are configured as a port for status messages.

More than one SFP+ port can be selected as an output for the status messages. In that case, the messages will be sent on both, with the same content, except for the TECMP counter (since this is specific for each output port).

5.3 System information tab

On the tab “System Information”, some status information about the device is displayed.

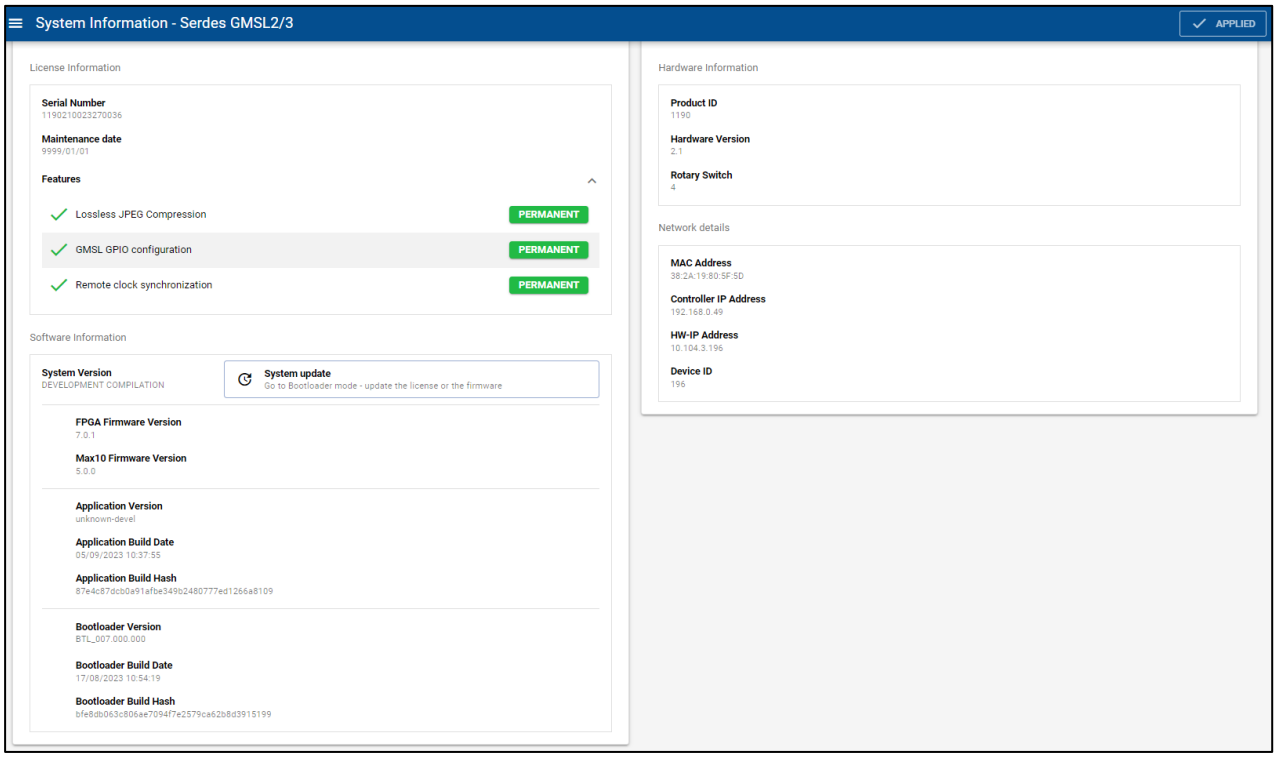


Figure 5-19 System Information Tab

5.3.1 License information

Serial number: Serial number that identifies the device unequivocally.

Maintenance date (DD-MM-YYYY): The date when the license is valid.

Features: The list of all the features that are covered under the license is mentioned here. There can be several modes of license.

1. Unlicensed: No license is available for this feature.
2. Temporal: The license is granted for the limited amount of time mentioned on the webpage.
3. Permanent: The license is granted permanently for this device.
4. Expired: The license for the feature mentioned is expired and the feature cannot be used anymore.

CM Serdes GMSL2/3 has the following features that need a license.

- Remote clock synchronization
- MIPI Datatype Overwrite
- Lossless JPEG Compression
- Adaptative Lossless Video Compression

5.3.2 Device information

Product Id: This shows the name of the device

Hardware version: The version of your hardware of the Capture Module Serdes GMSL2/3 is displayed here

MAC address: This MAC address should be the same as on the label at the bottom of the device.

Controller IP address: The user will be able to change the IP address of the Capture Module Serdes GMSL2/3, in this field. If the user wants to use multiple devices in one network, the user must configure a unique IP address for each device here. **This function is currently not supported by the software.**

HW IP address: IP address currently configured for the RJ45 port configured as configuration port.

Device ID: The TECMP Device ID is used for TECMP status and configuration messages.

5.3.3 Firmware Information:

System Update: This option will be used to update the firmware on the device.

⚠ When clicking on this option, the device goes into bootloader mode and loses its normal operation. The bootloader mode can also be verified by the fast toggle of the Host and FPGA status LED which in normal mode is in slow toggle mode. If this option is selected by mistake, please select “go back to application mode” to exit the bootloader mode.

MAX10 version: Additional FPGA needed for providing extended functionalities. The current version will be shown here.

FPGA version: It specifies the version used by the FPGA module. The current version will be shown here.

Application software version: Current firmware version installed on the microcontroller of the device.

Application build hash and date: This shows the internal reference to the repository build with the date.

Bootloader version: Current bootloader version of the microcontroller is used in the device.

Bootloader build hash and date: This shows the internal reference to the repository build with the date.

5.4 Control panel tab

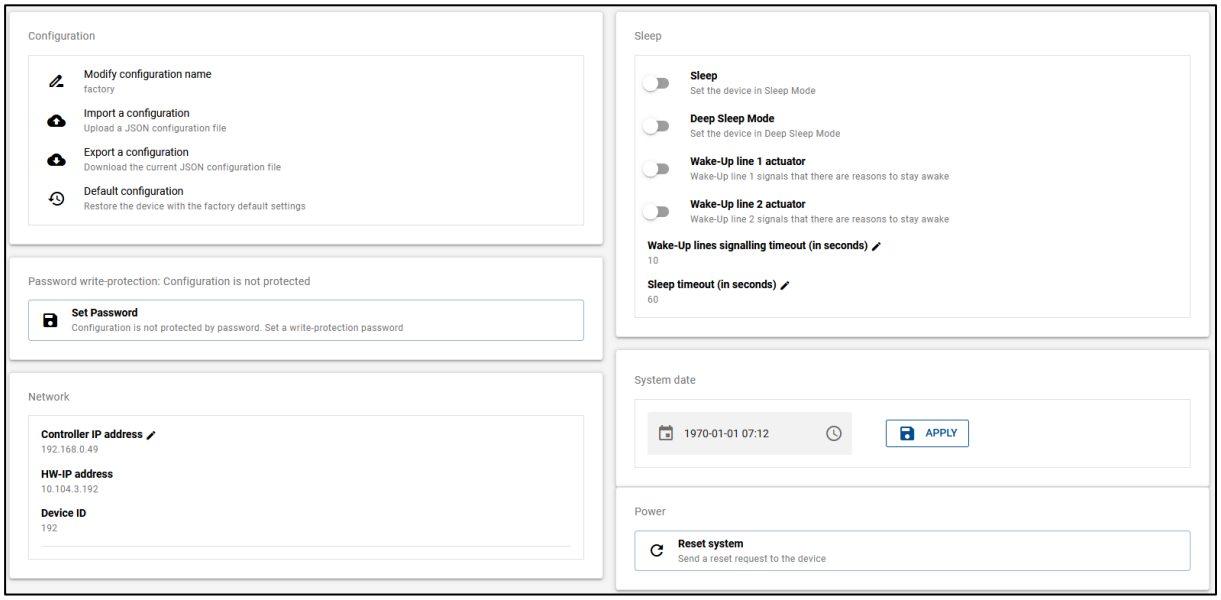


Figure 5-20 Control Panel Tab

5.4.1 Configuration

Users can make general settings of the configuration in the Capture Module Serdes GMSL2/3 in this tab

5.4.1.1 Import a configuration

Through this option, the user can import a configuration from the computer to the Capture Module Serdes GMSL2/3 in .json format.

The JSON file for configuration contains a section for the I2C advanced mode configuration. This file can be manually edited and then uploaded to the webpage to modify the I2C initialization sequence. An example of the file is shown in the image below:

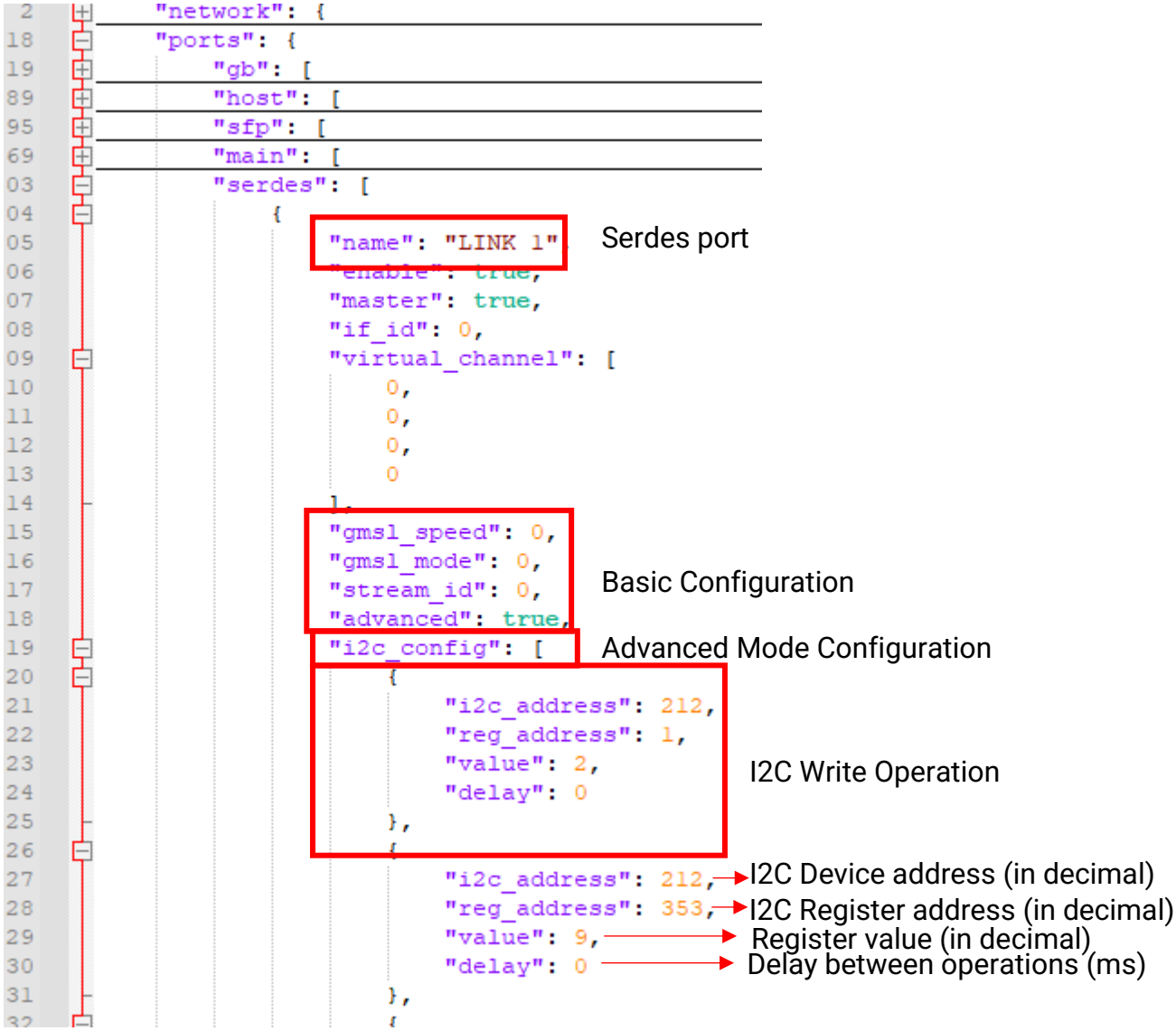


Figure 5-21 JSON format for I2C configuration

5.4.1.2 Export a configuration

Applied changes can be exported to the computer through this option. The exported configuration is in *.json file extension format.

5.4.1.3 Default configuration

Through this option, the Capture Module Serdes GMSL2/3 is reset to the default configuration. The IP address remains the same and is not reset to default.

5.4.2 Password write-protection

This function allows the user to set a password, ensuring that the configuration cannot be modified without authorization. Once a password is set, it will be required for making any changes to the device's configuration. If the password is forgotten, the only way to modify the configuration is to perform a full reset of the device to its factory settings.

5.4.3 System date

Date and Time: With this field, the time of the Capture Module can be set to be taken over as a reference for the timestamp in the TECMP protocol.

This setting can be useful in case the Capture Module Serdes GMSL2/3 is not connected to a time master, to have a timestamp close to real-time. For example, a standalone test bench with a gPTP Master. In this scenario, the timestamps are set based on the internal clock of the device and, if the *Date and Time* are not updated before making use of the device, timestamps will be calculated using the default date of 01.01.1970 00:00:00 as the basis.

To update the date and time of your device, select:

 To update the date: When  is selected, CM is set to the current date.

 To update the time: When  is selected, CM is set to the current time.

5.4.4 Power

This section offers settings for power and sleep functions.

5.4.5 Reset System

This option can be used for a soft reset of the device.

5.4.6 Network

This section offers settings about the network address and identifiers used for the Capture Module Serdes GMSL2/3.

5.4.6.1 Controller IP Address

IP address to access the Capture Module Serdes GMSL2/3 through the GB ports. The user will be able to change the IP address of the Capture Module Serdes GMSL2/3, in this field when the rotary switch is set to position “E”. If the user wants to use multiple devices in one network, the user must configure a unique IP address for each device here.

5.4.6.2 HW IP Address

IP address to access the Capture Module Serdes GMSL2/3 through the 100BaseT1 port. The user will be able to change the IP address of the Capture Module Serdes GMSL2/3, in this field when the rotary switch is set to position “E”. If the user wants to use multiple devices in one network, the user must configure a unique IP address for each device here. **This function is currently not supported by the software.**

5.4.6.3 Device ID

The Device ID is used for PLP status and configuration messages.

5.4.7 Sleep parameters

The Capture Module Serdes GMSL2/3 can work in two different operation modes:

Normal operation mode: The CM Serdes GMSL2/3 works at full capacity.

Sleep Mode: If required, it is possible to configure the device to enter the sleep mode using this option. By default, this option is not enabled.

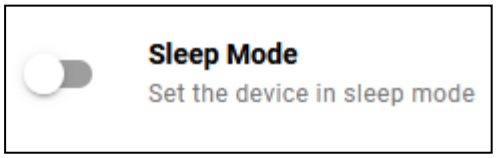


Figure 5-22 Prevent sleep



Not selected: The Capture Module Serdes GMSL2/3 works at full capacity and never enters the sleep mode



Selected: The Capture Module Serdes GMSL2/3 works at full capacity if there is communication on any of the monitored ports. If there is no communication for a defined time in the field SLEEP TIMEOUT (IN SECONDS) the CM Serdes GMSL2/3 goes into sleep mode.

Deep Sleep Mode: If required, it is possible to configure the device to enter the deep sleep mode using this option. When the CM Serdes GMSL2/3 is in deep sleep mode, it consumes 1mA. By default, this option is not enabled.

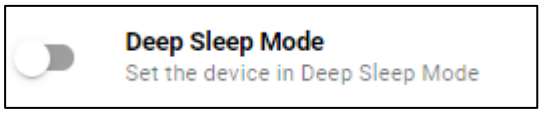


Figure 5-23 Prevent deep sleep

Not selected: The Capture Module Serdes GMSL2/3 works at full capacity and never enters the deep sleep mode

Selected: The Capture Module Serdes GMSL2/3 works at full capacity if there is communication on any of the monitored ports. If there is no communication for a defined time in the field SLEEP TIMEOUT (IN SECONDS) the CM Serdes GMSL2/3 goes into deep sleep mode.

5.4.7.1 Keep-alive conditions

All ports of the Capture Module Serdes GMSL2/3 - host included - can be monitored to detect traffic, to determine whether it is necessary to keep the device in normal operation, capturing data, or not.

If there is communication on any of the monitored ports or a high level on one of the wake-up lines of the device is detected, the CM Serdes GMSL2/3 will remain in a normal operation state.

As soon as all keep-alive reasons disappear and no wake-up line is activated, the device switches from normal operation to sleep mode, reducing the power consumption as described above.

There are two wake-up lines, which can be used anytime to wake up the CM Serdes GMSL2/3 in case it is configured to allow sleep mode, and the device enters this state.

5.4.8 Wake-up conditions

If the device is in sleep mode, it can only be woken up by activating one of the two wake-up lines.

A power reset will not wake up the device from sleep mode if it is off for less than 15 seconds. Only the wake-up lines will have an immediate effect if activated below those 15 seconds.

5.4.8.1 Wake-up lines and actuator

There are two wake-up lines, referred to as Wake-Up line 1 actuator and Wake-Up line 2 actuator. Both wake-up lines can work as input and output, speaking from the point of view of the device:

- Input (a.k.a. WUP source): Activation of the wake-up line from the outside will wake up the CM Serdes GMSL2/3.
- Output (a.k.a. WUP actuator): The CM Serdes GMSL2/3 can activate the wake-up line to keep other devices connected to it awake or wake them up if they were in sleep mode.

Using the wake-up lines as input line is always possible and cannot be deactivated. Using them as output lines is configurable and can be activated/deactivated independently for each line using the setting *Wake-Up Line X actuator*, as described below:

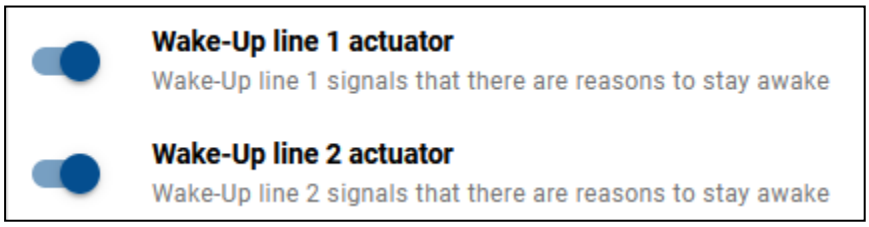


Figure 5-24: Wake-up lines



Wake-Up line X actuator selected: Wake-Up line X actuator is used also as output line and the corresponding pin (WUP*X*) provides the same voltage (vs. GND) as the supply voltage of the Capture Module if the device is awake.

Wake-Up line X actuator not selected: Wake-Up line X actuator is only input and its activation from the outside is used only to wakes up the device if it was previously in sleep mode.

By default, both wake-up lines are disabled as the actuator.



In case a wake-up line is configured as actuator - thus, it can work both as input and as output line – and the line should be at the same time controlled in both directions, in case of discrepancy (one tries to activate it, the other one to deactivate it), the line will be activated.

5.4.8.2 Wake-Up lines signaling timeout

This setting defines the time in seconds that must pass by before the wake-up lines if configured as actuators – according to *Wake-Up line X actuator* –, are deactivated after the reasons to keep the device awake have disappeared. This timer starts counting as soon as no wake-up and no keep-alive reason is present anymore.

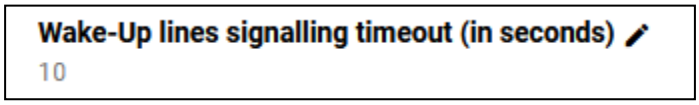


Figure 5-25: Wake up line signaling timeout

This setting makes only sense when at least one of the wake-up lines is configured as an actuator. In case none is configured, both lines should be already deactivated (since there is no wake-up reason present) and thus no timeout is required.

The default value for this setting is 10 seconds.

5.4.8.3 Sleep Timeout

This setting determines how much time must pass by with no communication on the monitored ports before the CM Serdes GMSL2/3 enters the sleep mode.

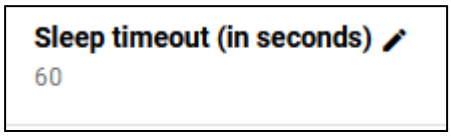


Figure 5-26: Sleep Timeout

Reasonable values for this setting are above 15s so that the user can get access to the webpage of the device.

If the field is left blank (i.e., not configured), the counter will start from power-up with default values.

To know how much time exactly the CM Serdes GMSL2/3 will require to go to sleep. In the case at least one of the wake-up lines is configured as an actuator, the total amount of time required for the device to enter the sleep mode will be the time for the wake-up line to be deactivated plus the sleep timeout.

5.5 Compression

On the tab “Compression“, the configuration for the video compressor can be uploaded. The compressor configuration can be modified for each individual GMSL link by uploading a JPEG or JPEG Header file (containing the Huffman tables and compression mode) to the drag and drop box .
Once the compressor configuration has been successfully uploaded, a reset of the CM Serdes GMSL2/3 is performed automatically to apply this configuration.

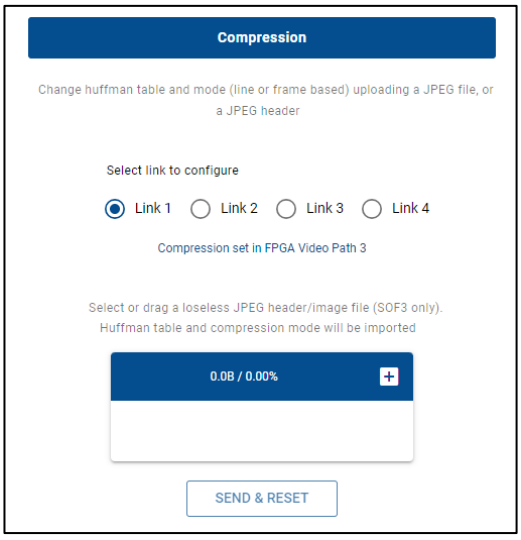


Figure 5-27 Compression Tab

5.6 Serdes Diagnostics

On the tab “Serdes Diagnostics”, arbitrary I2C operations (read and write) can be performed to any GMSL link control channel. This allows for configuring and reading the status of the different devices on the link.

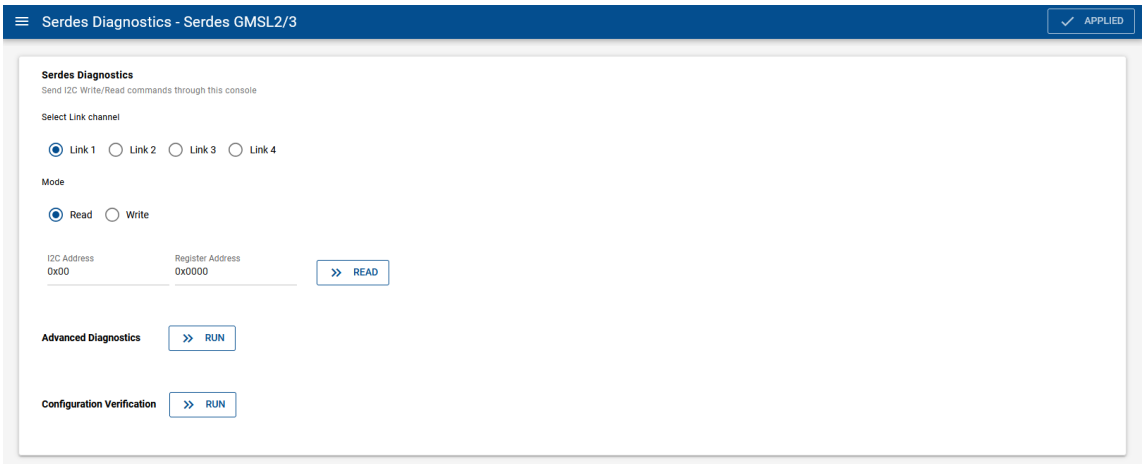


Figure 5-28 Serdes Diagnostics Tab

The Serdes diagnostics tab can be used for **PRBS testing** by configuring the serializer and deserializer devices on PRBS mode (following Maxim guidelines defined in the GMSL2 user manual) by requesting I2C read and write operations through the serial console.

Advanced Diagnostics

There is also the possibility to run a predefined set of advanced diagnostics to read relevant control registers. These diagnostics provide a clearer view of the SerDes link

behavior, enabling users to quickly identify and resolve issues to ensure reliable operation.

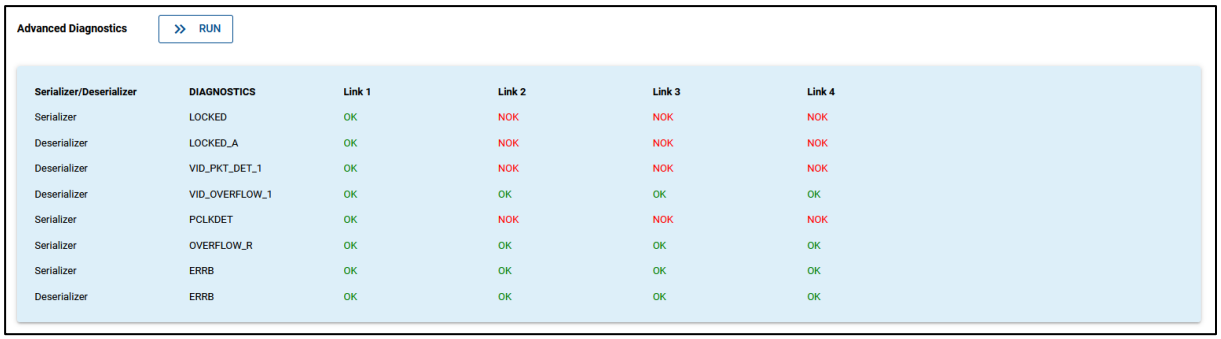


Figure 5-29 Advanced Diagnostics

Configuration Verification

A configuration verification is available to detect potential SerDes link misconfigurations, identifying the specific bit in error along with its expected value. To resolve the problem, the user may perform a device reset.

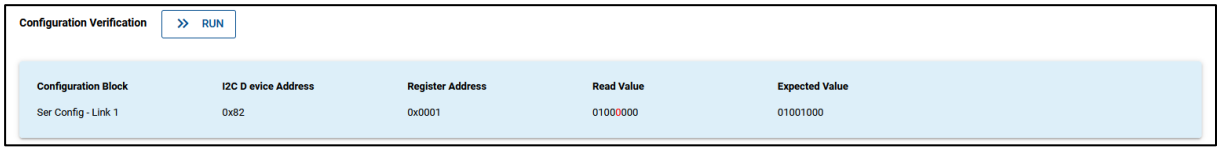


Figure 5-30 Configuration Verification

6. Maintenance

6.1 Firmware update

This section explains how a Capture Module can be upgraded from a release to higher releases.

6.1.1 Preconditions and important information

If an error occurs during the update, please perform a power reset and try again

Make sure your network adapter is configured in the correct IP address range

The device can only be flashed/updated through the GB-A port

Please follow these update instructions to avoid erroneous states of the device. Technica Engineering may charge support fees for repair services.

A stable 12 Volt DC power supply is required. Do not switch off the power supply during the update process

6.1.1.1 Update via GB-A port

Pre-requisites:

- Please set the device to default settings

Steps:

1. Open the website through the configuration port and by typing the IP of the device defined by the rotary switch.
2. Go to the “System Information” tab and click System update

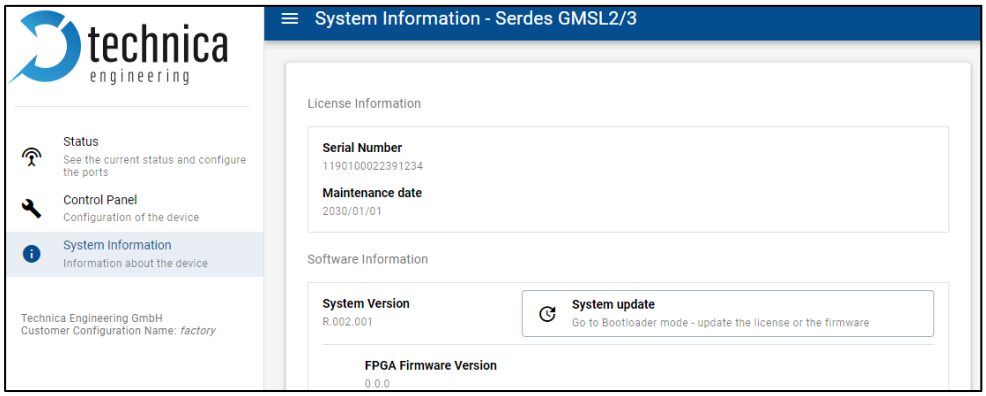


Figure 6-1 System information tab

- The device is now in bootloader mode and the following website should appear:

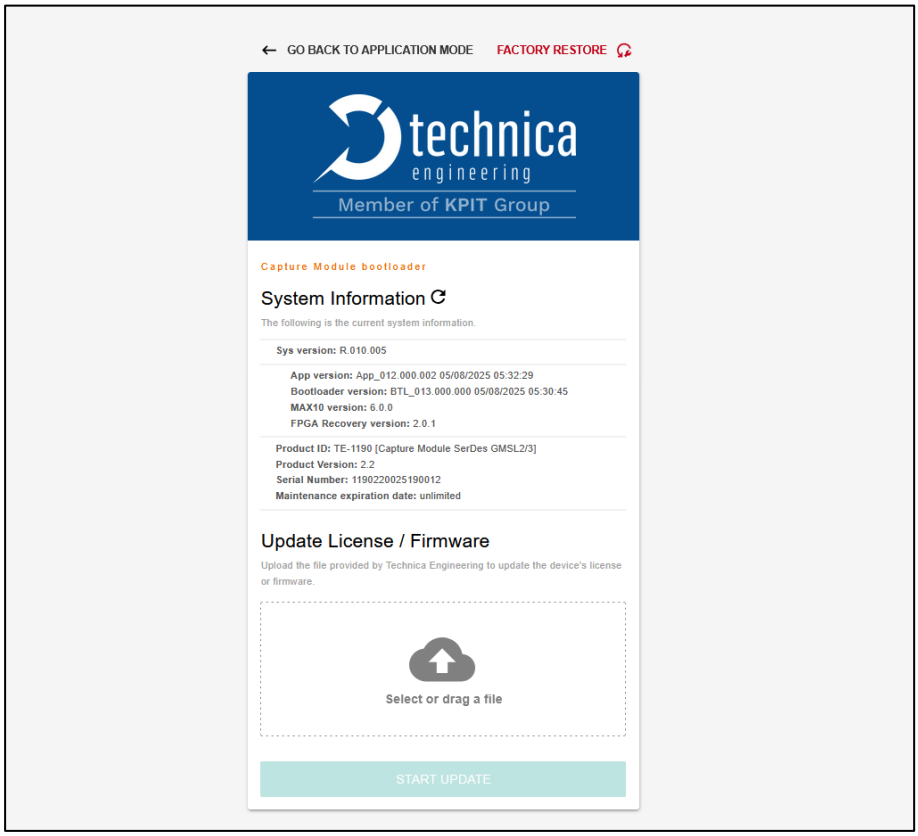


Figure 6-2 Firmware upload

3. Please click into the framed field and open the file XXXX.TEFW for new firmware or open new license files. Alternatively, it is possible to drag and drop the mentioned files int to the framed field
4. Press START UPDATE 

5. Wait until you are prompted by the following window:

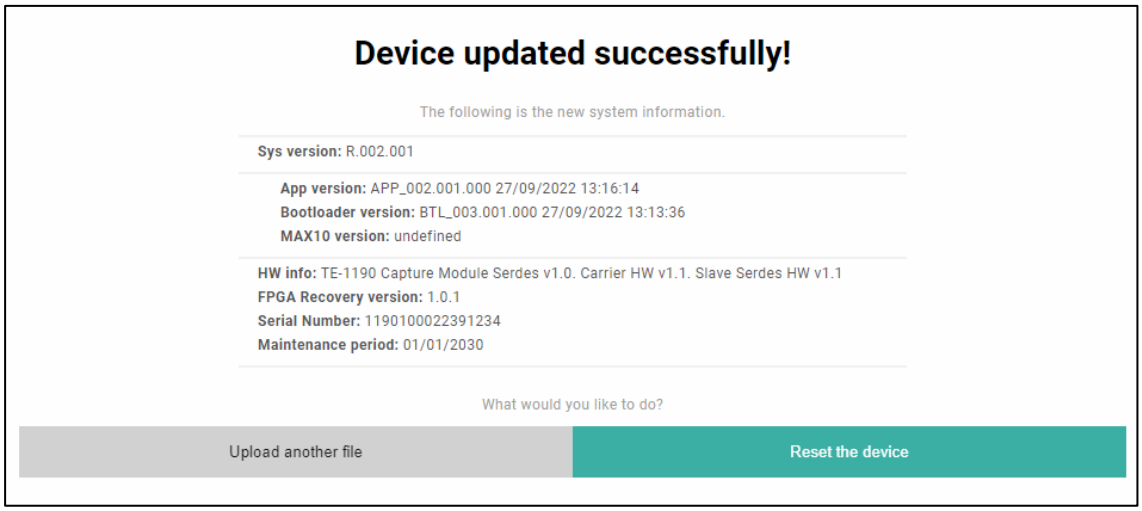


Figure 6-3 Firmware update successful

6. Press “Reset the device” if you are finished with updating.
7. The device will jump into application mode and show you how the website is again configured.

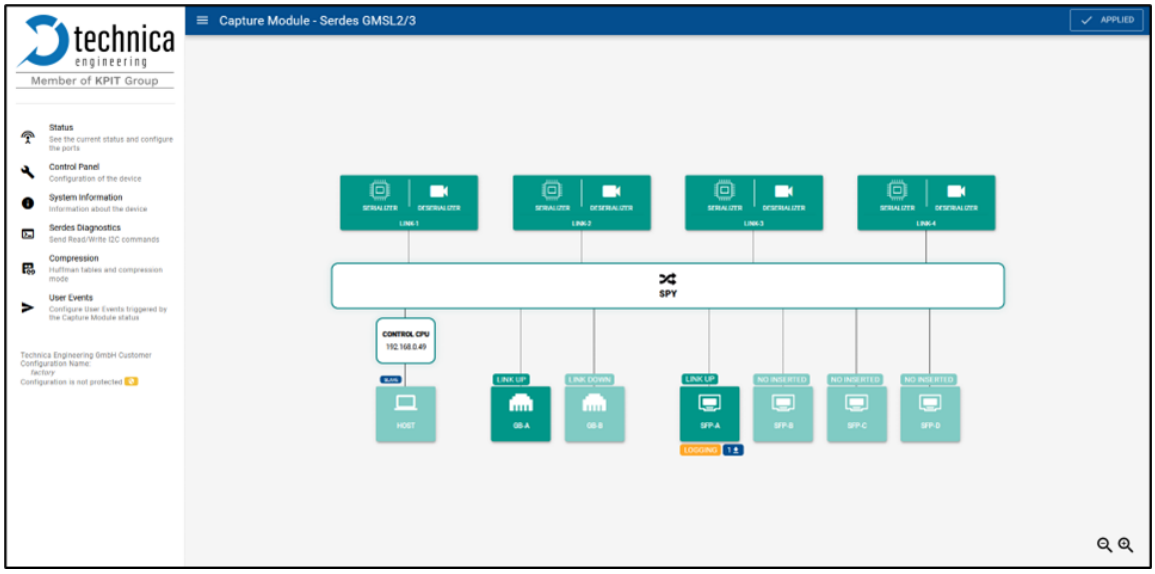


Figure 6-4 Home screen

➔ Finished

7. PLP Logging

7.1 PLP MIPI Packet

The Capture Module will log all MIPI CSI-2 packets sent over each GMSL 2/3 bus following the PLP MIPI standard to the selected SFP+ port. The BusSpecID of the packet will correspond to the one configured on the website by the user depending on the MIPI CSI-2 Virtual Channel and GMSL 2/3 link.

- BusSpecID according to virtual channel config

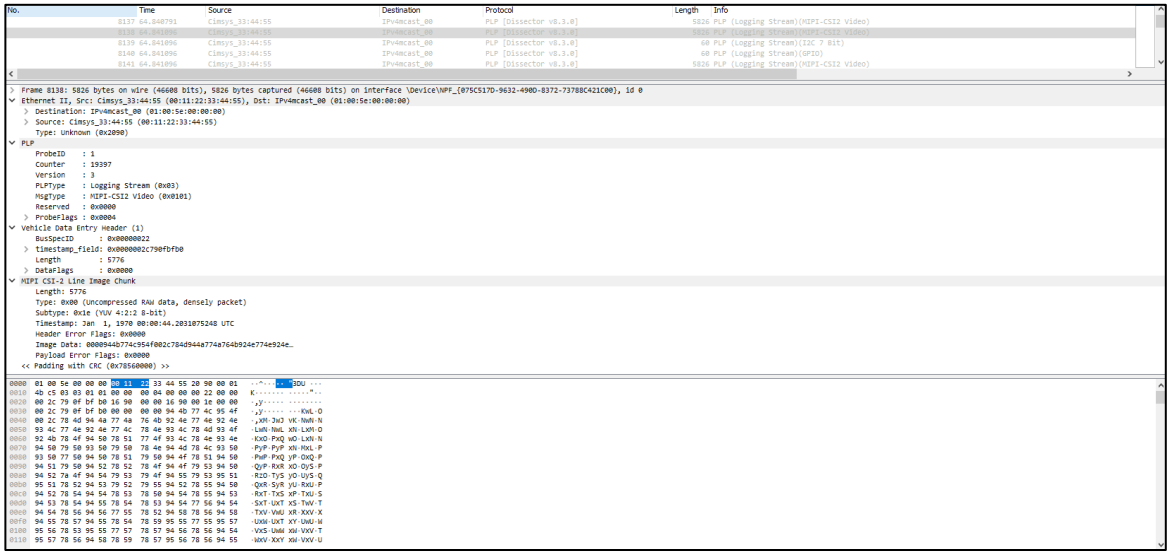


Figure 7-1 Captured MIPI Packet

7.2 PLP I2C Packet

The Capture Module will log all I2C transactions sent over each GMSL 2/3 link following the I2C PLP standard to the selected SFP+ port. The BusSpecID of the packet will correspond to the one configured in the website by the user depending on the GMSL 2/3 link.

- BusSpecID according to Link config

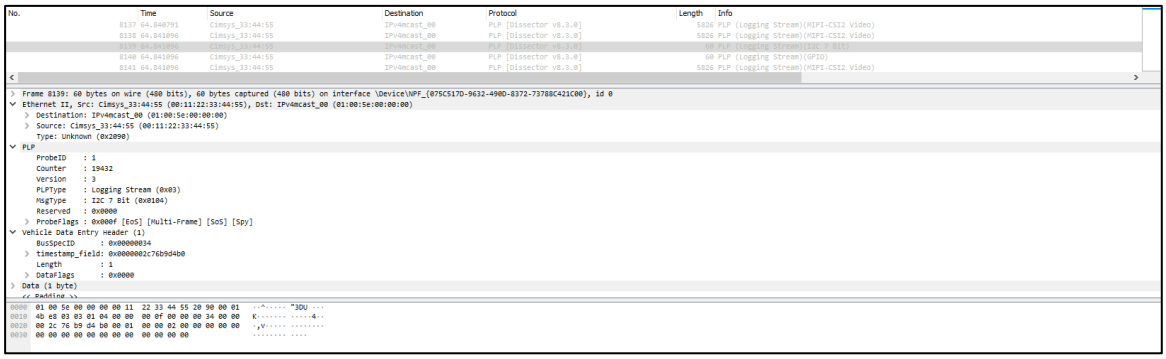


Figure 7-2 Captured I2C Packet

7.3 PLP GPIO Packet

The Capture Module will log all GPIO transactions sent over each GMSL 2/3 link following the GPIO PLP standard to the selected SFP+ port.

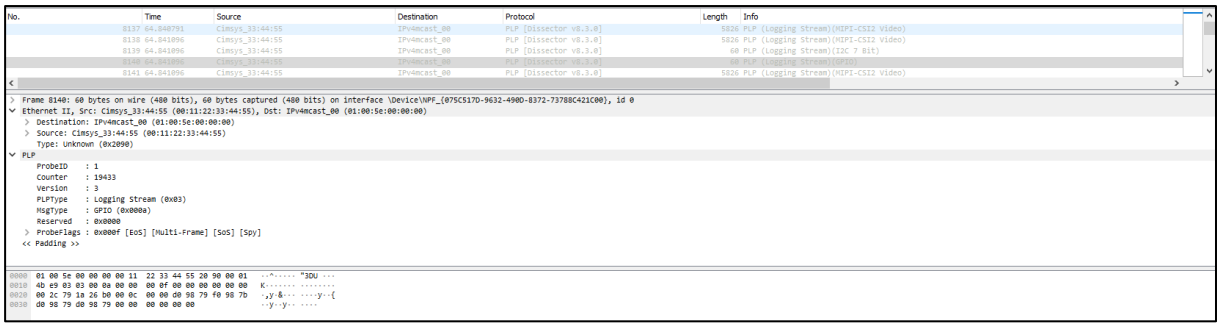


Figure 7-3 Captured GPIO Packet

8. Changelog

Version	Chapter	Description	Date
1.1	All	Initial version	Oct 2022
1.2	All	Updates	Oct 2022
1.3	All	Updates for new release	Mar 2023
1.6	All	FPGA video path and SPY configuration added	Nov 2023
1.7	All	Adaptative compression and selective DataType Overwrite added	Apr 2024
1.8	All	Updated document for product version 3.0	Oct 2025

9. Contact

If you have any questions regarding this product, please feel free to contact us:

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Latest user manuals and product information:
https://technica-engineering.de/en/